

Bio 30 – Unit A Review

Your Exam

▶ What to expect?

▶ 45 MC 7 NR 15 marks of WR

▶ Diploma Style/Context questions

- ▶ Most questions are context-based, meaning they require reading.
- ▶ Often times totally new biological contexts will be covered, however the material learned in Bio 30 will allow you to answer these questions
- ▶ Ex: A question refers to ALS (neurodegenerative diseases). You won't need to know anything about ALS it will be explained in the question. The question itself will be from the Biology 30 curriculum

Studying Tips

- ▶ **Active Review!**
- ▶ Summarize
 - ▶ Condense each chapter into as small as possible (1 page max each)
- ▶ Practice tests
 - ▶ Self-assessment, Exam Bank, Quest A+
- ▶ Flash cards/Quizlet, etc.
- ▶ Visualize and connect
 - ▶ Notes in your own words, making connections, webs and drawings

Unit A - Outcome 1

Nervous System

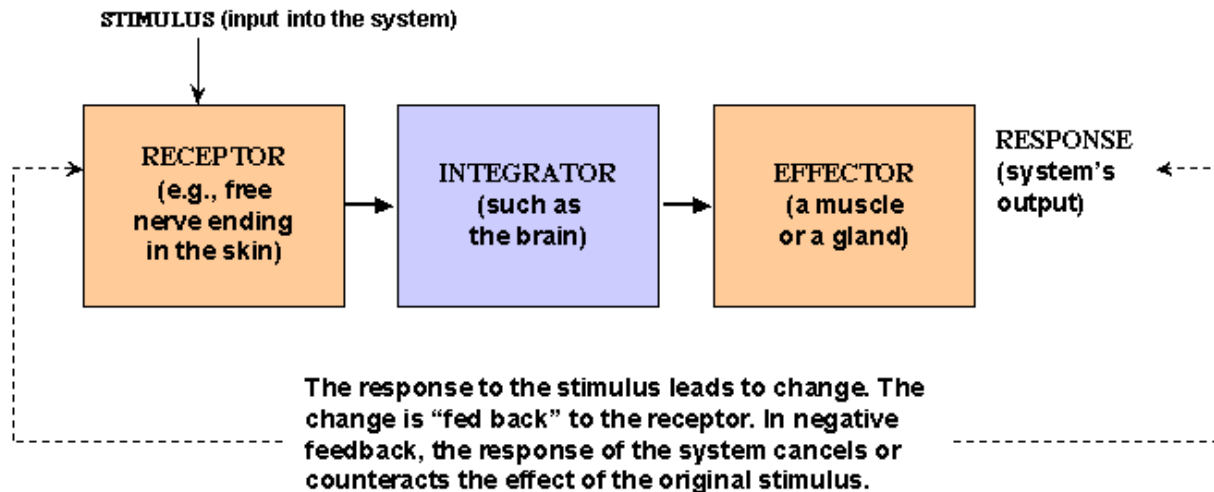
- ▶ describe the general structure and function of a neuron and myelin sheath, explaining the formation and transmission of an action potential, including all-or-none response and intensity of response; the transmission of a signal across a synapse; and the main chemicals and transmitters involved, i.e., norepinephrine, acetylcholine and cholinesterase
- ▶ identify the principal structures of the central and peripheral nervous systems and explain their functions in regulating the voluntary (somatic) and involuntary (autonomic) systems of the human organism; i.e., cerebral hemispheres and lobes, cerebellum, pons, medulla oblongata, hypothalamus, spinal cord, sympathetic and parasympathetic nervous systems, and the sensory-somatic nervous system
- ▶ describe, using an example, the organization of neurons into nerves and the composition and function of reflex arcs; *e.g., the patellar reflex, the pupillary reflex*
- ▶ describe the structure and function of the parts of the human eye; i.e., the cornea, lens, sclera, choroid, retina, rods and cones, fovea centralis, pupil, iris and optic nerve
- ▶ describe the structure and function of the parts of the human ear, including the pinna, auditory canal, tympanum, ossicles, cochlea, organ of Corti, auditory nerve, semicircular canals and Eustachian tube
- ▶ explain other ways that humans sense their environment and their spatial orientation in it; *e.g., olfactory receptors, proprioceptors, taste receptors, receptors in the skin.*
- ▶ observe the principal features of a mammalian brain, eye and ear, and identify the major structures of those organs

Homeostasis

- ▶ The human body works to maintain homeostasis, which is a constant internal condition. Nervous and endocrine systems work towards this

Homeostasis

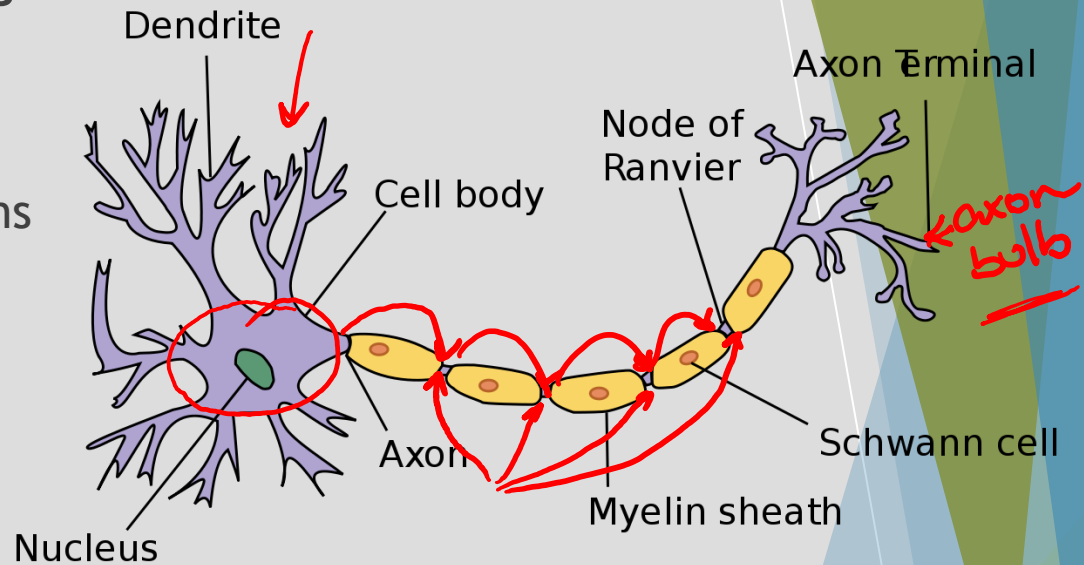
- Stable operating conditions in the internal environment
- Three components interact



What is a neuron (nerve cell)

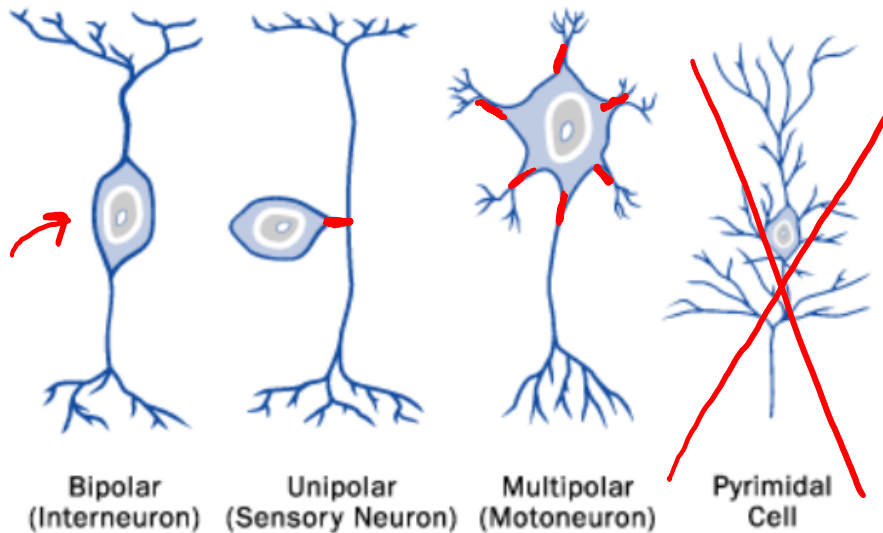
- ▶ Neuron = nerve cell that transmit information (A nerve is a bundle of neurons)

- ▶ Cell Body - contain the nucleus and two extensions
- ▶ Dendrites - shorter (look like tentacles) receive information
- ▶ Axons - single long 'fiber' which conducts nerve impulse, carries information
- ▶ Schwann cells - make up myelin sheathe, speed up impulse by allowing action potential to jump to each node of ranvier



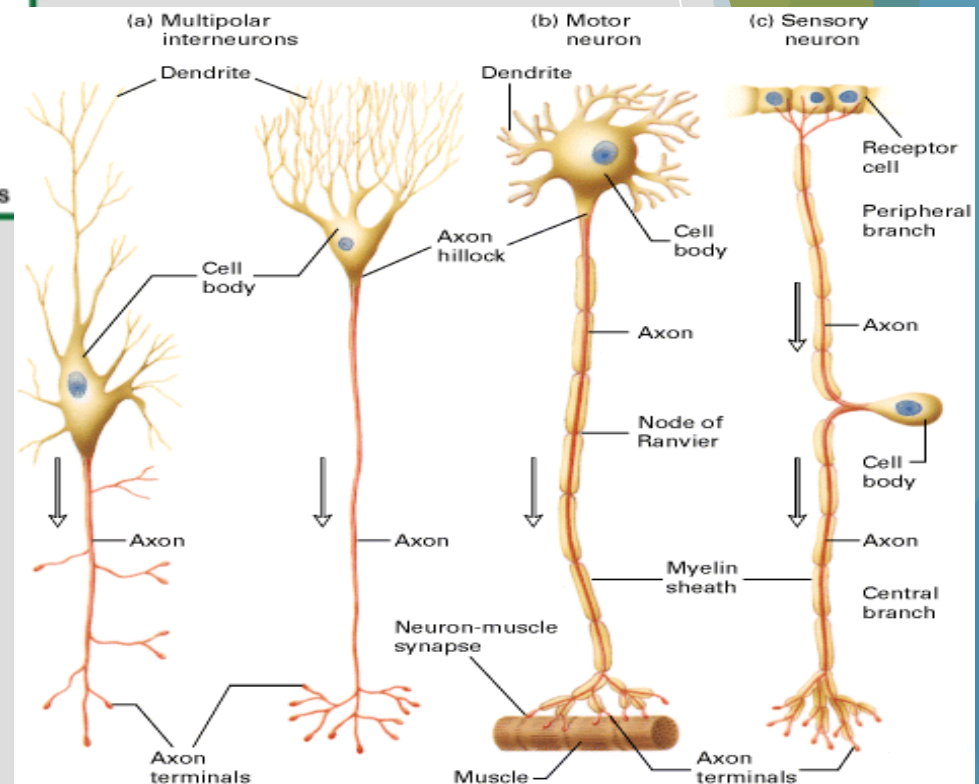
Neuron types

Basic Neuron Types



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- ▶ A) Bipolar - interneurons
- ▶ B) Unipolar - sensory neurons
- ▶ C) Multipolar - Motor neurons

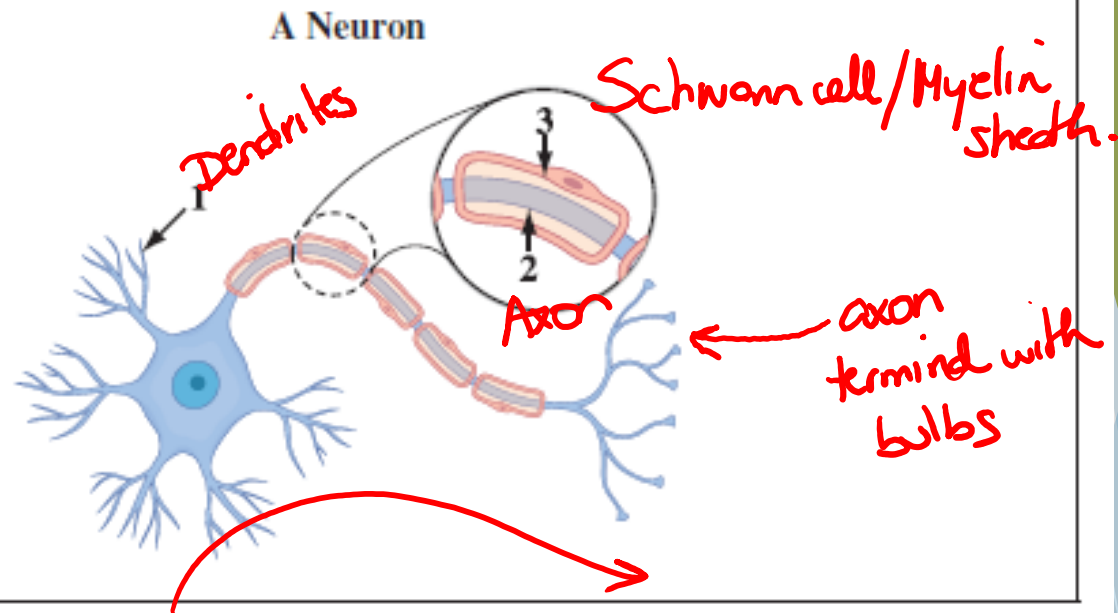


► Schwann cells (white matter)

- [illegible]

Use the following information to answer question 1.

Charcot-Marie-Tooth disease (CMT) is a neurological disorder characterized by nerve damage that causes muscle weakness and loss of sensation. The two most common forms of this disorder are CMT1 and CMT2. CMT1 is caused by a genetic mutation that results in abnormal structure and function of the myelin sheath, and CMT2 is caused by a genetic mutation that leads to the abnormal functioning of the axon.



1. Which of the following rows identifies the structures of a neuron affected by CMT1 and CMT2? ←

Row	CMT1	CMT2
A.	1	2
B.	2	3
C.	3	2
D.	3	1

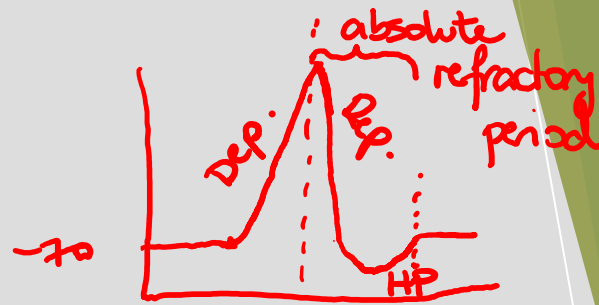
Diploma Question

Neurofibromatosis is a genetic disorder characterized by the growth of tumours called neurofibromas around some neurons. Neurofibromas result from the uncontrolled growth of cells that produce the myelin sheath.

3. The structures that are affected by neurofibromas are the

- A. axons of neurons in grey matter
- ☒ B. axons of neurons in white matter
- ~~C.~~ dendrites of neurons in grey matter
- ~~D.~~ dendrites of neurons in white matter

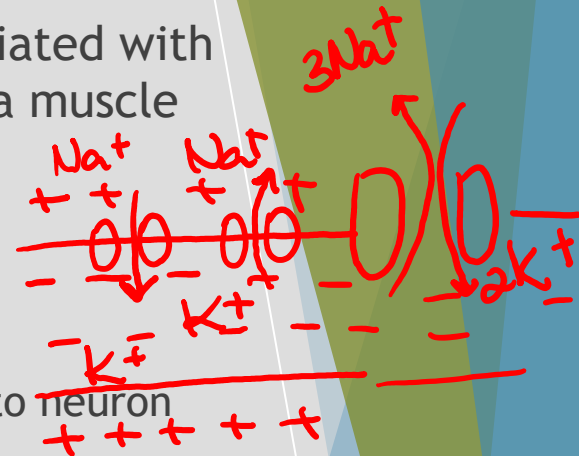
Action Potentials



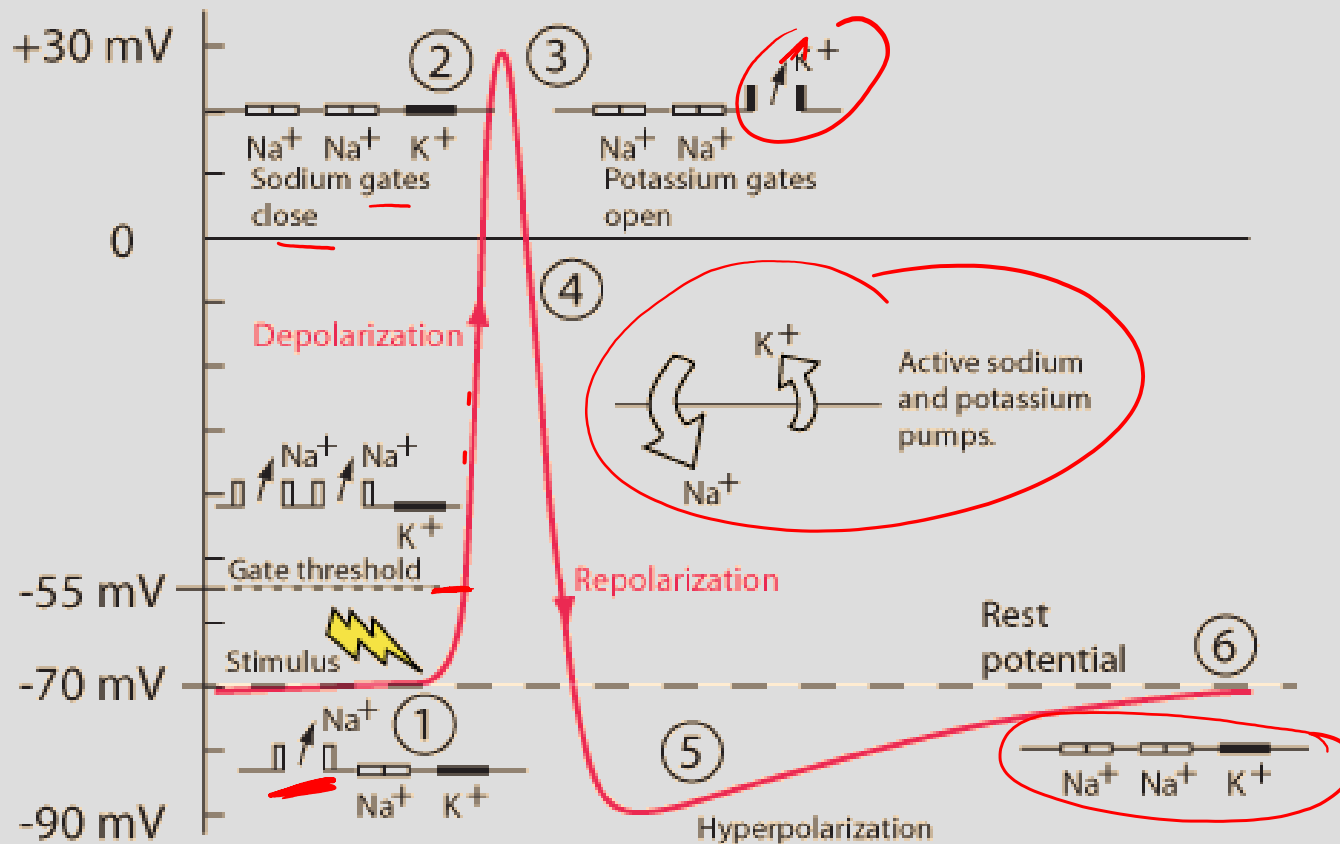
- ▶ Definition: the change in electrical potential associated with the passage of an impulse along the membrane of a muscle cell or nerve cell

- ▶ Steps:

- ▶ Stimulus received (must pass threshold potential)
- ▶ Depolarization: sodium gates open, sodium flows into neuron (makes inside positive)
- ▶ Sodium gates close and potassium open
- ▶ Repolarization: potassium flows out of neuron, makes it negative again
- ▶ Hyperpolarization: repolarization step overshoots resting potential, important to make sure impulse moves in one direction
- ▶ Rest potential brought back by sodium-potassium pump (uses ATP)



Action potentials



<https://www.youtube.com/watch?v=b2ctEsGEpe0>
<https://www.youtube.com/watch?v=oa6rvUJlg7o>

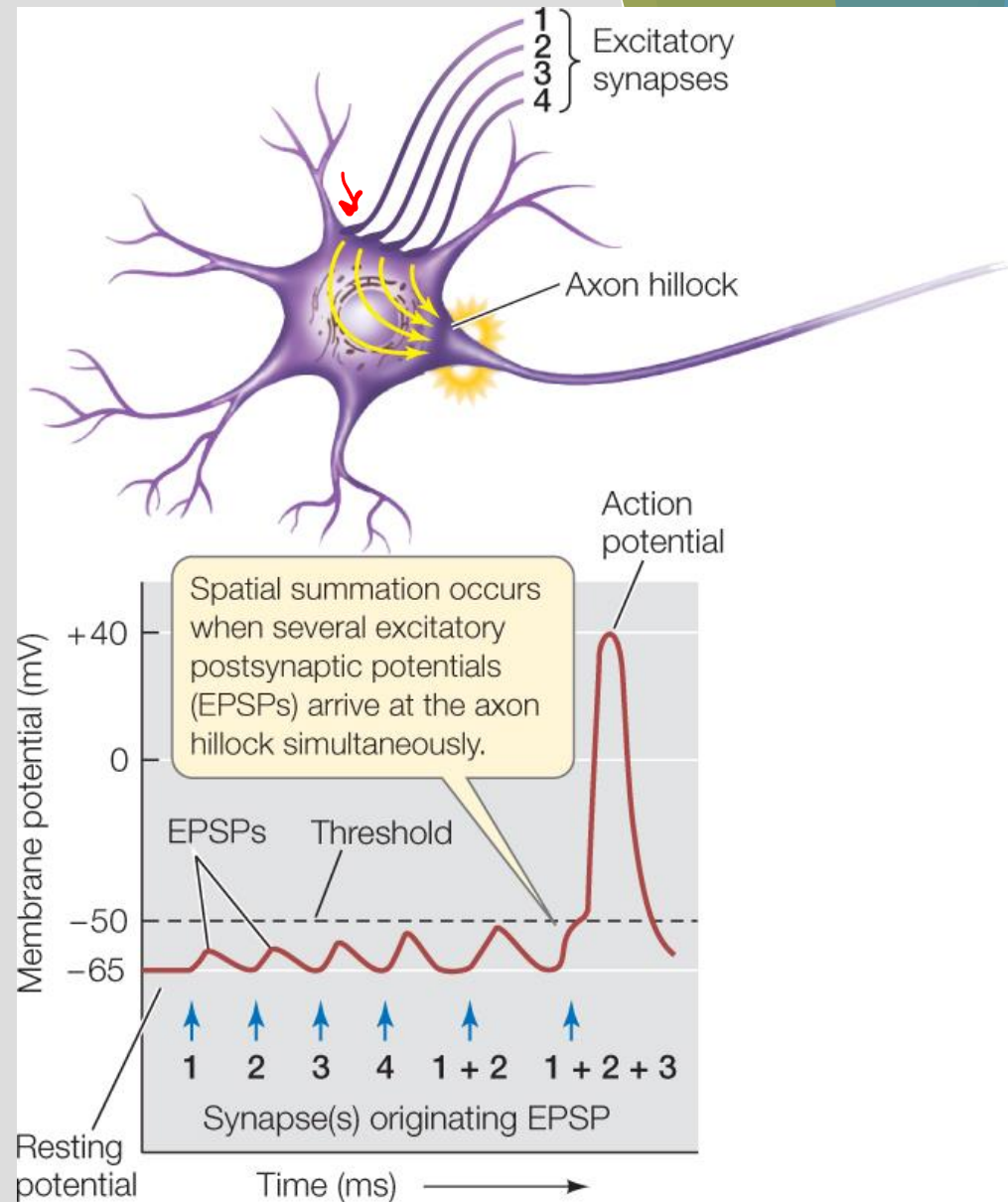
Action potentials



- ▶ All or none response: the action potential either happens or does not happen. There is no such thing as a 'weak' or 'strong' action potential.
- ▶ Intensity of response: depends on the frequency of action potential

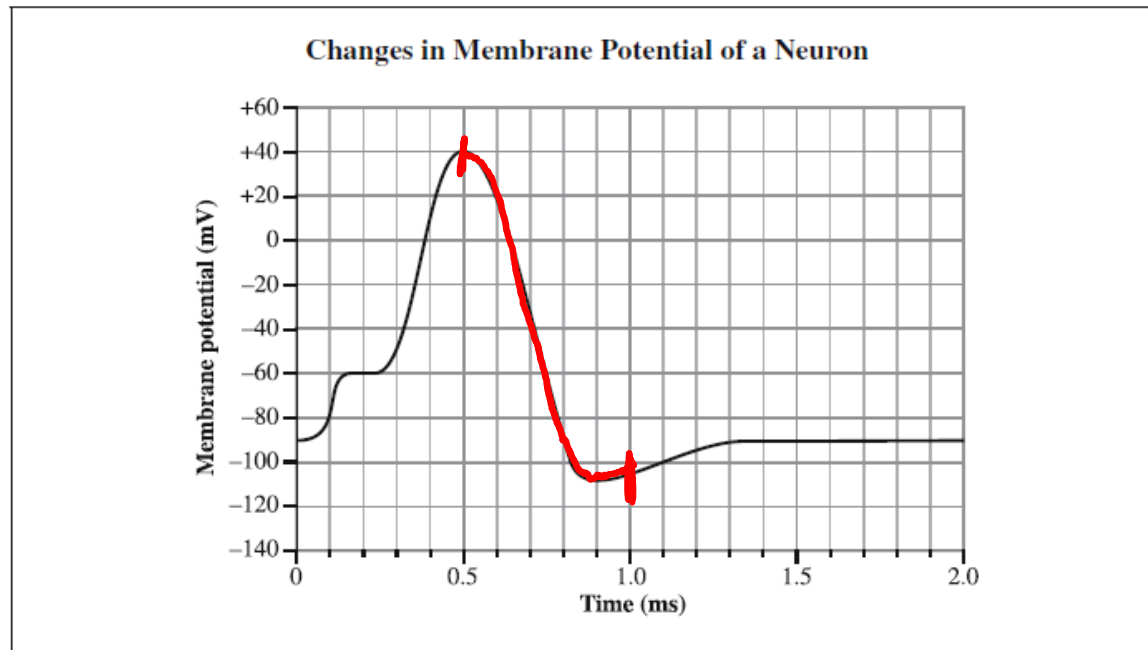
Summation

- ▶ Summation is the process that determines whether or not an action potential will be generated by the combined effects of **excitatory** and **inhibitory** signals
- ▶ may or may not reach the threshold voltage to trigger an action potential.
 - ▶ Spatial summation: from multiple simultaneous inputs
 - ▶ Temporal summation: repeated inputs



Practice

Use the following information to answer numerical-response question 2 and questions 5 and 6.



Numerical Response

2. What is the resting membrane potential for the neuron represented in the graph above, expressed as two digits, and what is the maximum membrane potential during depolarization, expressed as two digits?

Number:
Membrane
Potential:

9 0 4 0
Resting Maximum During
Potential: Depolarization

Practice

5. In the graph on the previous page, the ion movement that occurs between 0.3 ms and 0.5 ms is best described as
- ☒ A. sodium ions moving into the axon
 - ☐ B. sodium ions moving out of the axon
 - ☐ C. potassium ions moving into the axon
 - ☐ D. potassium ions moving out of the axon
6. On the graph above, the part of an action potential that occurs between 0.5 ms and 1.0 ms is best described as the
- ☒ A. refractory period, which is when repolarization occurs
 - ☐ B. refractory period, which is when depolarization occurs
 - ☐ C. threshold period, which is when repolarization occurs
 - ☐ D. threshold period, which is when depolarization occurs

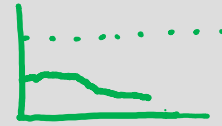
Use the following information to answer question 6.

A stimulus from each of two neurons is required to open some receptor channels in a brain neuron.

6. Which of the following aspects of nerve-impulse transmission is exemplified by the combined action of two neurons that causes stimulation of the brain neuron receptors?

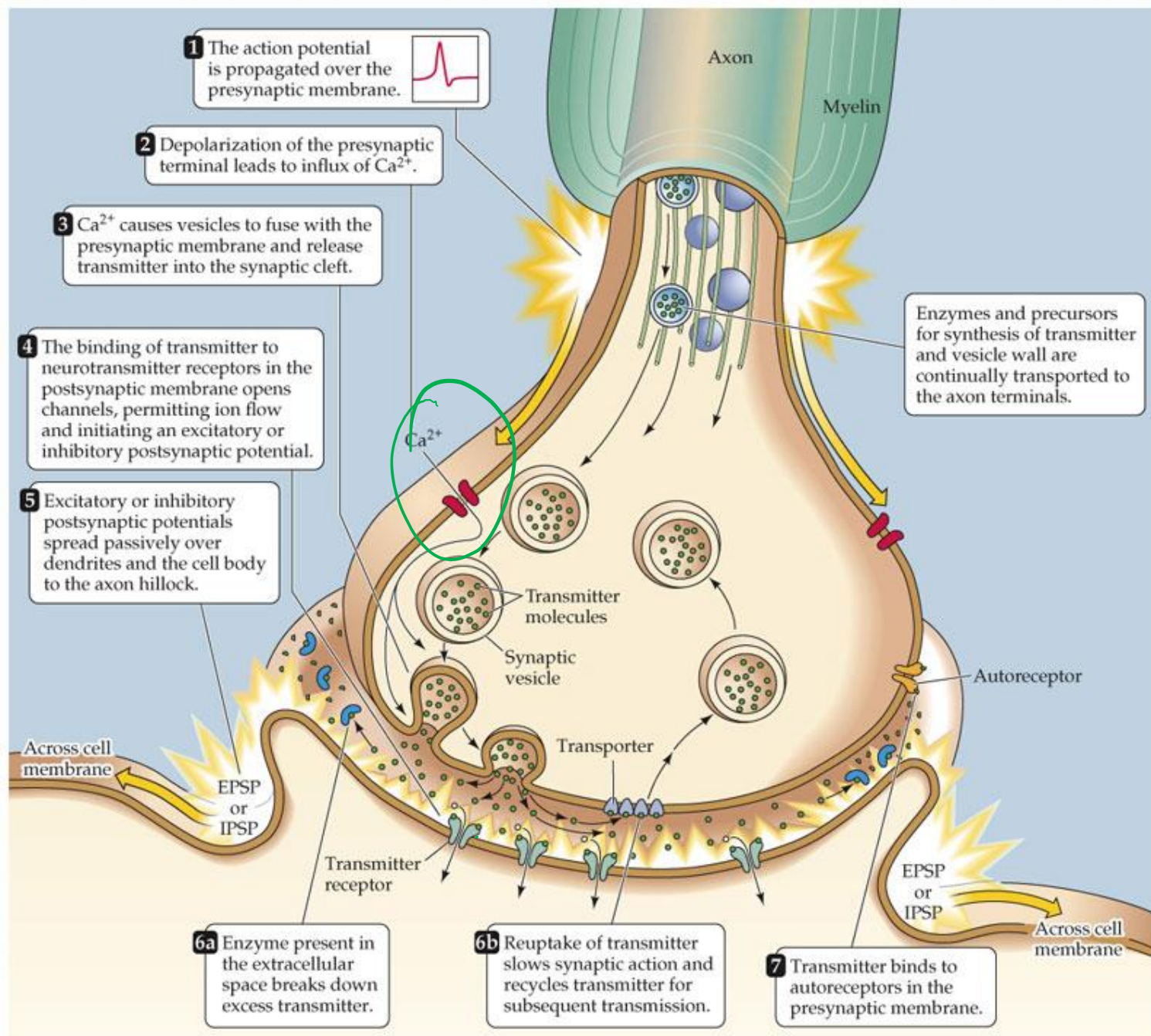
- A. Summation
- B. Depolarization
- C. Refractory period
- D. All-or-none response

Synapses



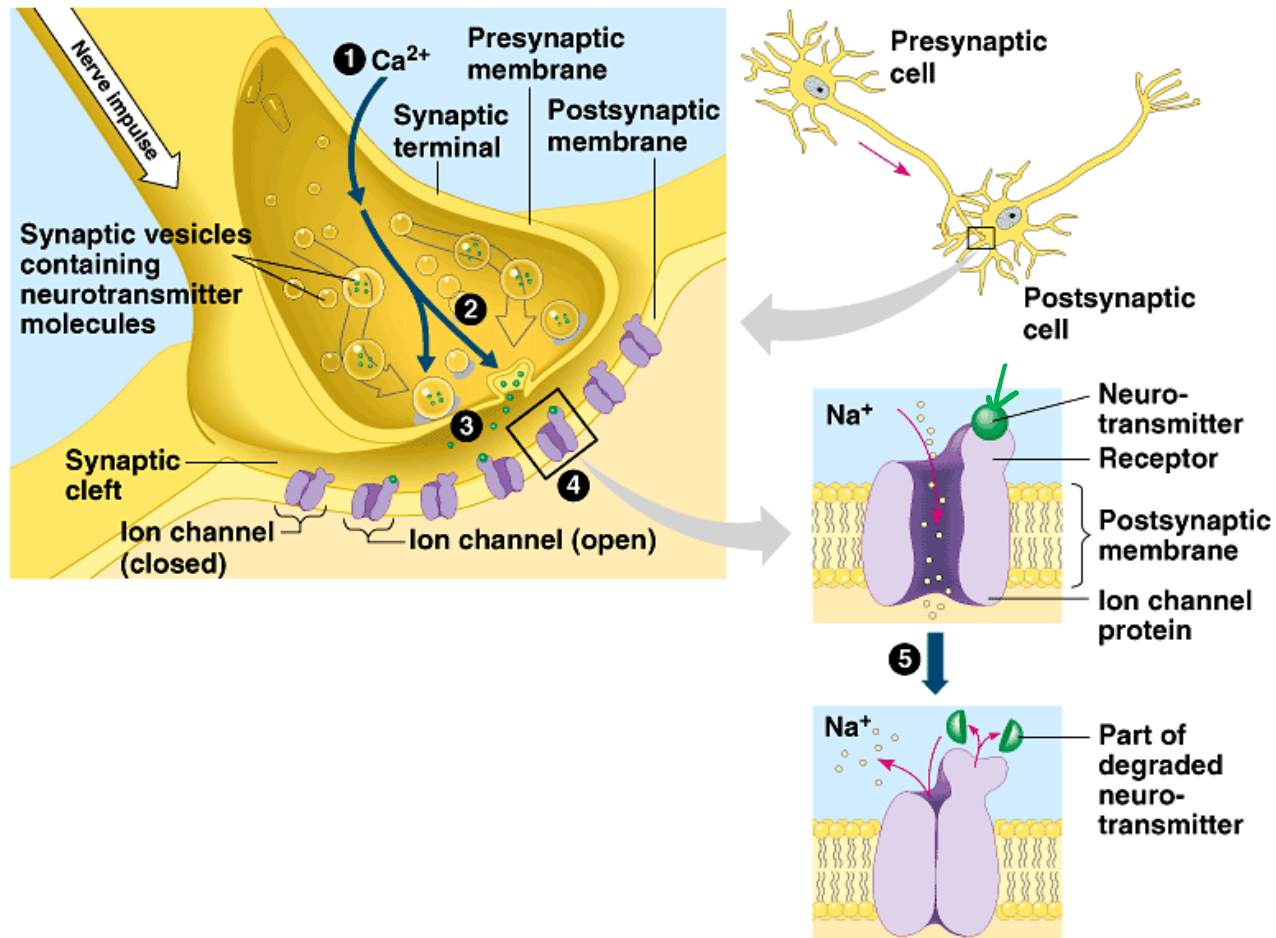
- ▶ How do neurons send signals to each other, since there is a gap (synapse) between?
 - ▶ Action potential reaches axon terminal
 - ▶ Causes calcium gates to open and Ca^{2+} flows in
 - ▶ Calcium fuses with vesicles full of neurotransmitters, these fuse to the membrane
 - ▶ Neurotransmitters released into synapses, diffuse across
 - ▶ Attach to post synaptic membrane and cause an effect
 - ▶ Excitatory - cause action potential (Na⁺ in)
 - ▶ Inhibitory - raise threshold potential
 - ▶ Release from receptors and are brought back into presynaptic neurons or are broken down by enzymes





THE MIND'S MACHINE 2e, Figure 3.11

Synapses



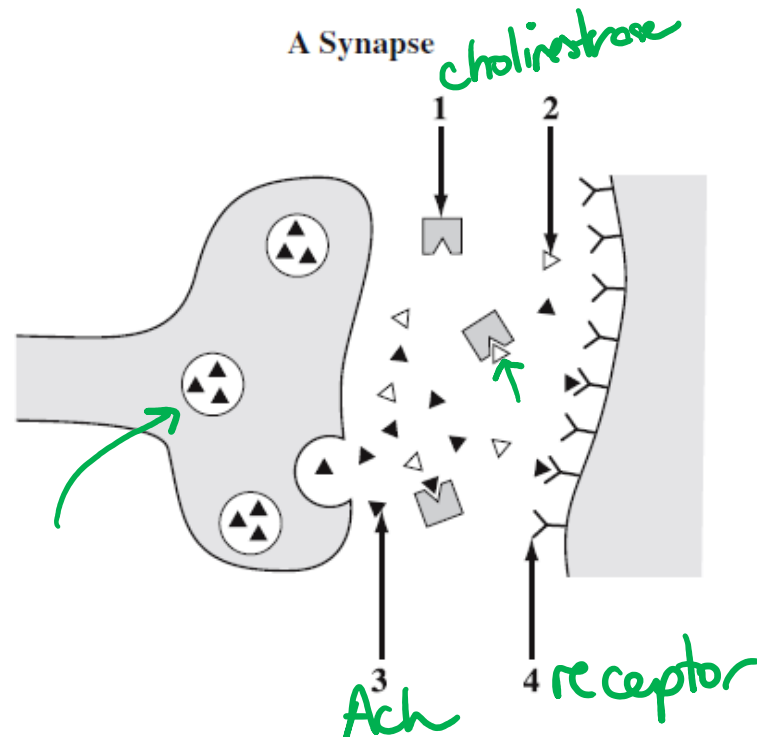
Synapses

- a. Simulation
 - a. <https://learn.genetics.utah.edu/content/addiction/mouse/>
- b. Neurotransmitters:
 - a. Excitatory
 - i. Ex. Glutamate, acetylcholine, epinephrine
 - b. Inhibitory
 - i. Ex. GABA
 - c. Enzymes
 - i. Cholinesterase

Practice

Use the following information to answer questions 1 and 2.

People with Alzheimer disease have a lower-than-normal level of acetylcholine in the brain. Cholinesterase inhibitors, such as the drug donepezil, can slow the development of symptoms in the early-to-middle stages of Alzheimer disease, but they cannot stop the progression of the disease. The donepezil molecule has a shape that allows it to attach to the active site on cholinesterase.



1. In the diagram above, donepezil and acetylcholine are numbered, respectively,

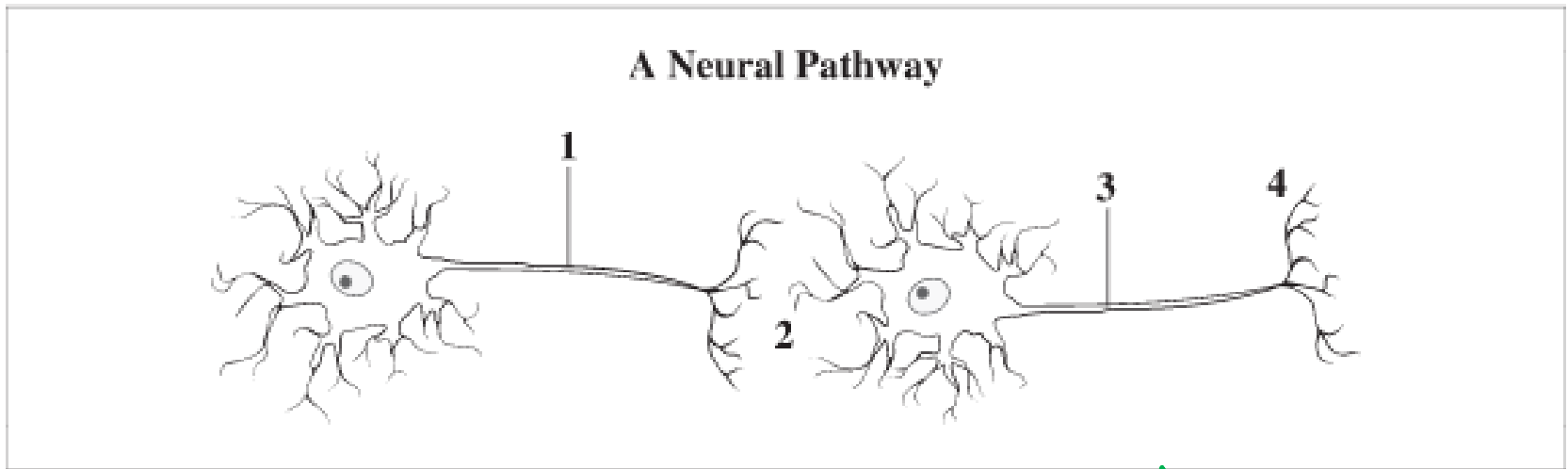
- A. 1 and 3
- B. 1 and 4
- ☒ C. 2 and 3
- D. 2 and 4

Practice

2. Which of the following statements describes an effect of donepezil on synaptic transmission?
- ~~A.~~ Donepezil breaks down acetylcholine so that less acetylcholine is present in the synapse.
 - ~~B.~~ Donepezil replaces cholinesterase so that more acetylcholine is present in the synapse.
 - ~~C.~~ Donepezil blocks the release of acetylcholine so that less acetylcholine is present in the synapse.
 - ☒ D. Donepezil prevents the breakdown of acetylcholine so that more acetylcholine is present in the synapse.

Practice

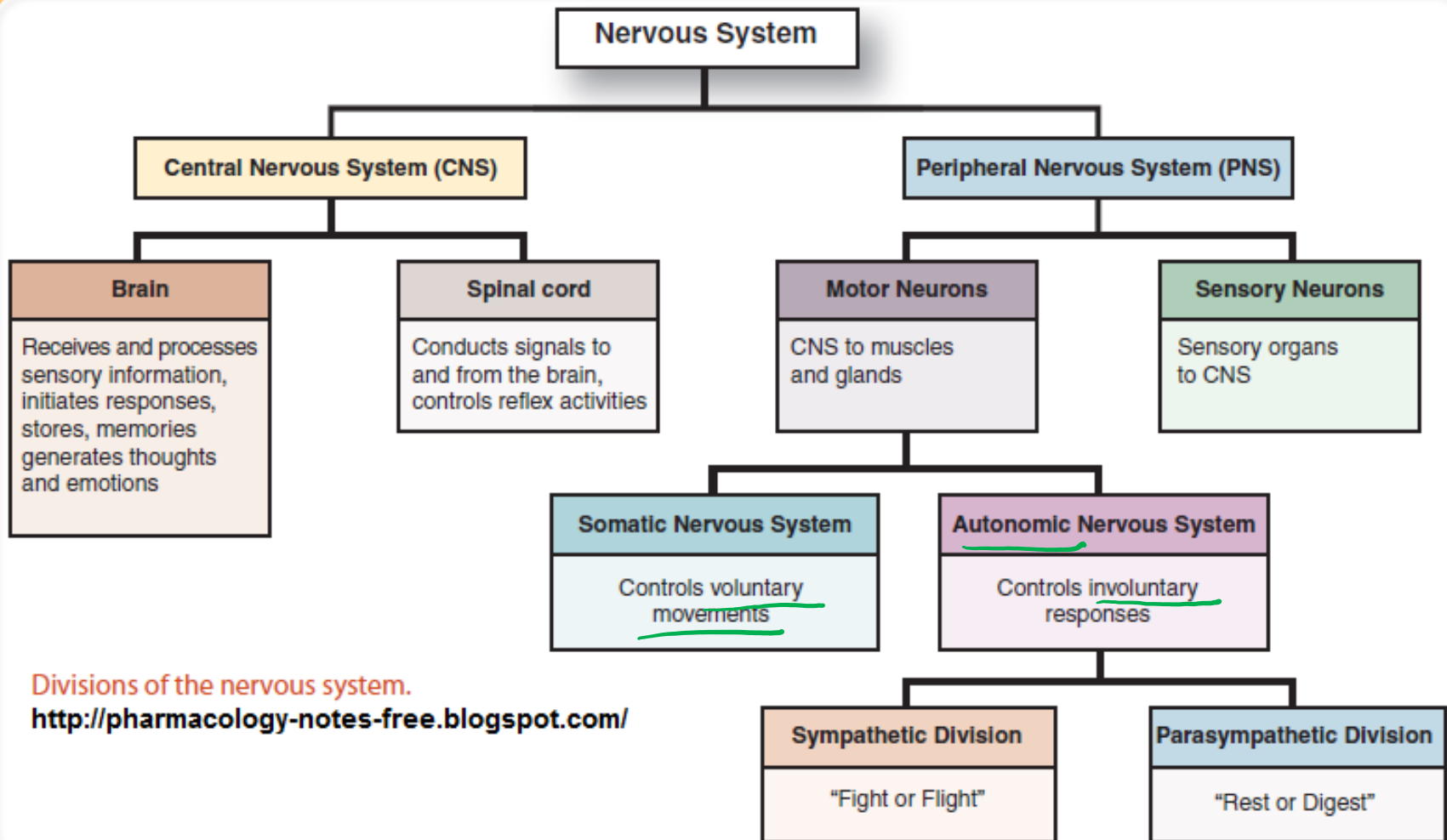
Use the following information to answer question 3.



3. In the diagram, the location in the neural pathway where the neurotransmitter dopamine functions is numbered

- A. 1
- ☒ B. 2
- C. 3
- D. 4

Divisions of nervous system



Divisions of the nervous system.

<http://pharmacology-notes-free.blogspot.com/>

Divisions of nervous system

- ▶ **Sympathetic vs parasympathetic, what would 'turn up' in a stressful situation? Brain storm:

→ ↑ HR

pupils dilate

↓ digestion

↑ blood pressure and flow to muscles and important organs and brain

↑ blood glucose

↑ breathing rate

Practice

4. Which of the following rows describes the change in heart rate, blood flow to the skin, and glucose storage that would be expected immediately following stimulation of the sympathetic nervous system?

stress

Row	Heart Rate	Blood Flow to Skin	Glucose Storage
☒ A.	Increased	Decreased	Decreased
☐ B.	Increased	Increased	Increased
☐ C.	Decreased	Decreased	Decreased
☐ D.	Decreased	Increased	Increased

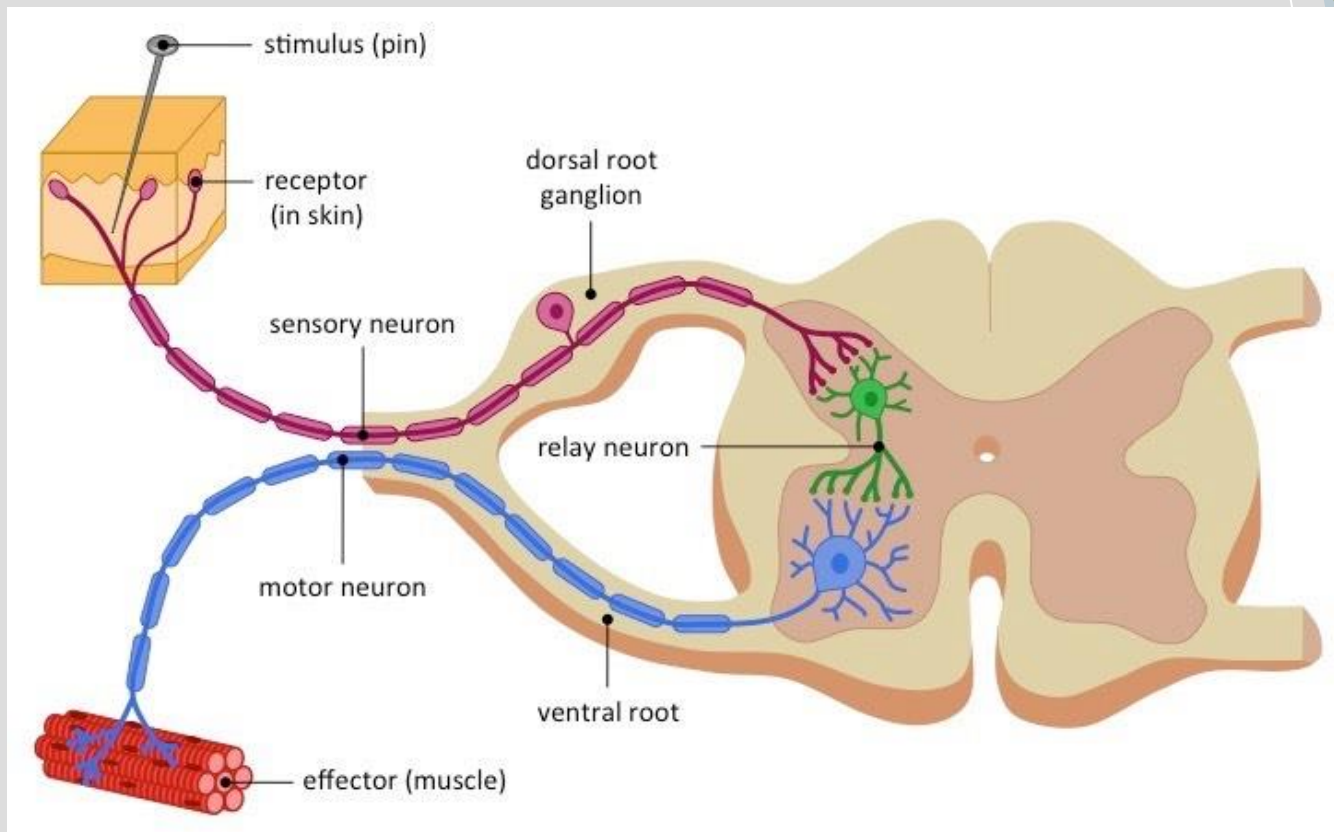
8. The division of the nervous system that is **directly** responsible for physiological responses to fear is the

- A. sensory nervous system
- B. somatic nervous system
- ☒ C. sympathetic nervous system
- D. parasympathetic nervous system

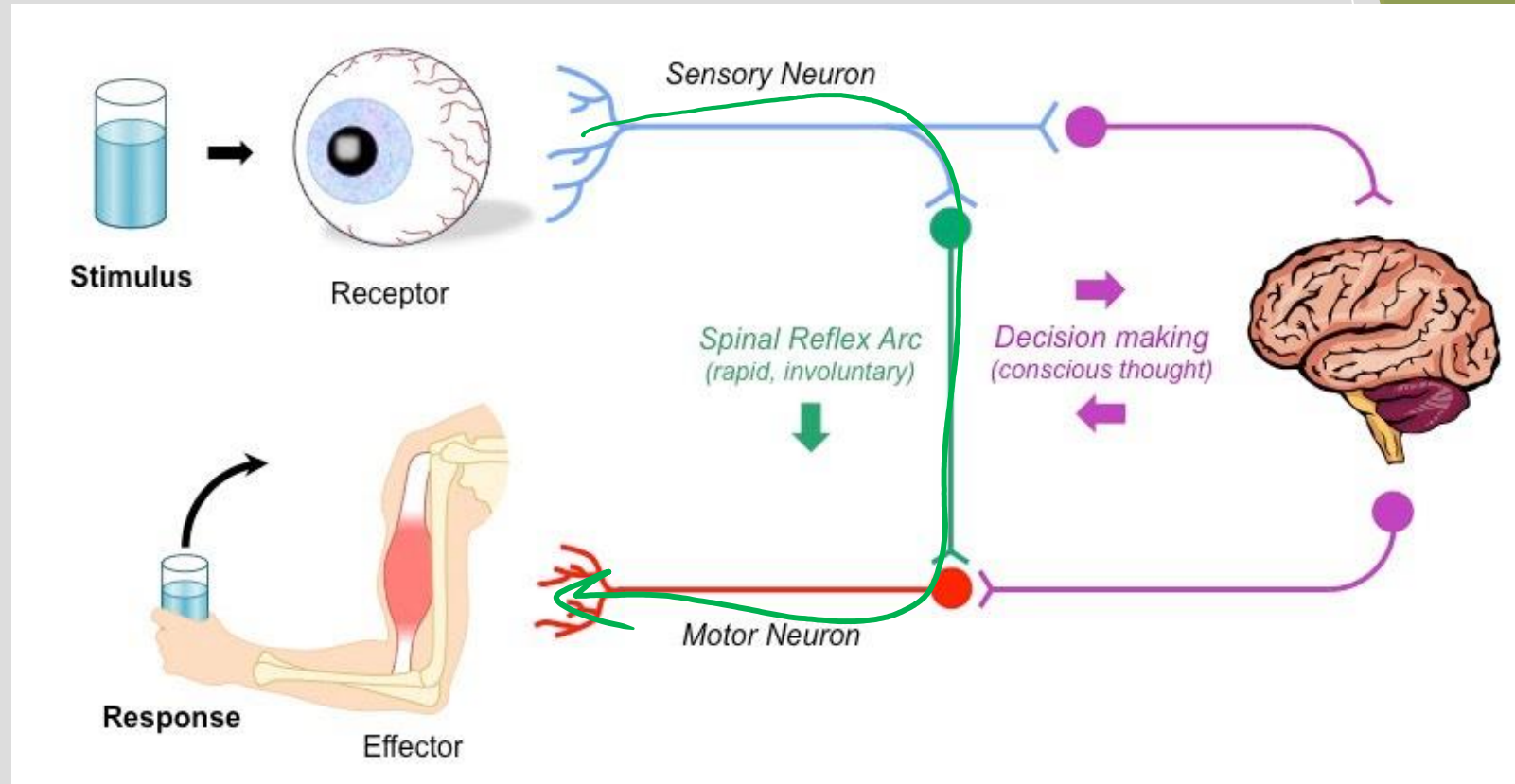
Reflex arcs

receptor → sensory neuron → inter neuron → motor neuron → effector

- Definition: the nerve pathway involved in a reflex action including at its simplest a sensory nerve and a motor nerve with a synapse between



Reflex arc



Practice

Use the following information to answer question 13.

After a man accidentally hits his thumb with a hammer, he initially does not feel pain, but immediately withdraws his hand.

13. Which of the following descriptions **best** explains the reason the man initially does not feel pain?
- A. The threshold of the receptors in the man's thumb has been so greatly exceeded that they do not pass the message to the brain.
 - B. There are too many nerve impulses travelling to the brain that the brain is unable to interpret the signal fast enough.
 - ☒ C. Neural processing occurred in the spinal cord first, which caused the man to quickly remove his thumb from further damage.
 - D. Sensory receptors in the man's thumb were damaged by the hammer and were unable to initiate a stimulus to the sensory nerve.

Practice

14. The correct order of structures involved in a reflex arc is
- A. receptor, sensory neuron, effector, motor neuron
 - B. motor neuron, interneuron, sensory neuron, effector
 - C. sensory neuron, receptor, interneuron, motor neuron
 - ☒ D. receptor, sensory neuron, interneuron, motor neuron

Use the following information to answer question 4 and numerical-response question 1.

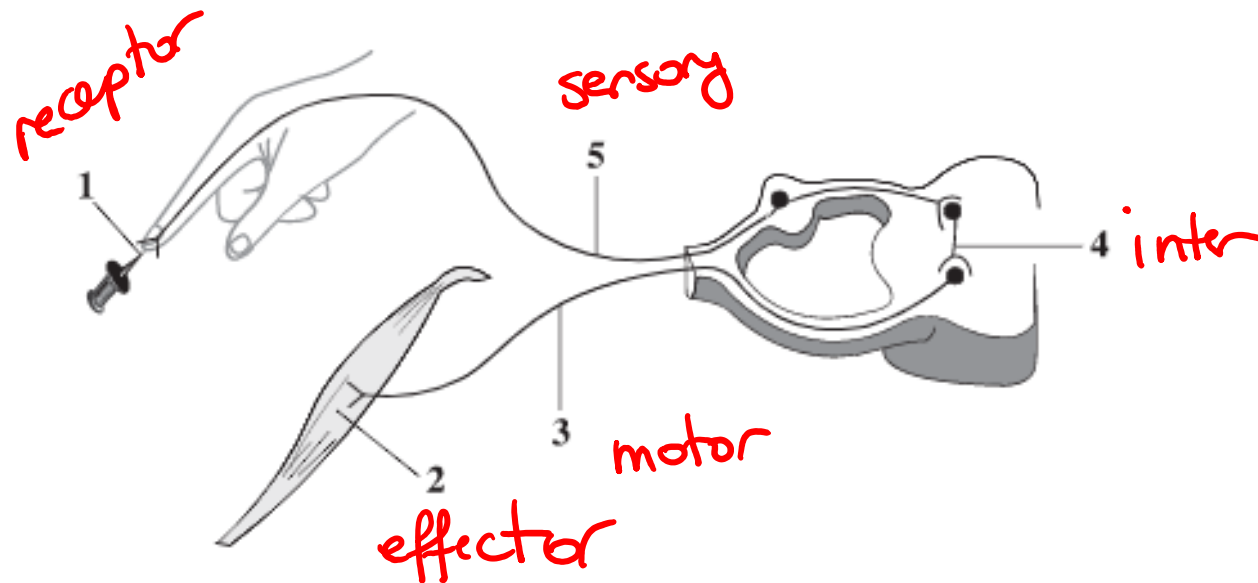
Hansen disease is caused by the bacterium *Mycobacterium leprae*. The bacteria attach to Schwann cells, which are wrapped around neurons, and damage the protein that provides a covering for the neurons.

4. Compared with the nerve impulses of neurons that have an undamaged covering, the nerve impulses of neurons of a person with Hansen disease are initially
- ☒ A. transmitted more slowly
 - ☐ B. transmitted continuously
 - ☐ C. transmitted more quickly
 - ☐ D. not transmitted across synapses

Use the following additional information to answer numerical-response question 1.

If Hansen disease is not treated, permanent damage to Schwann cells results. People who have the disease often injure their hands and feet because they do not sense the stimuli that would normally help to warn them of harm, and their withdrawal reflex does not occur.

A Reflex Arc



Numerical Response

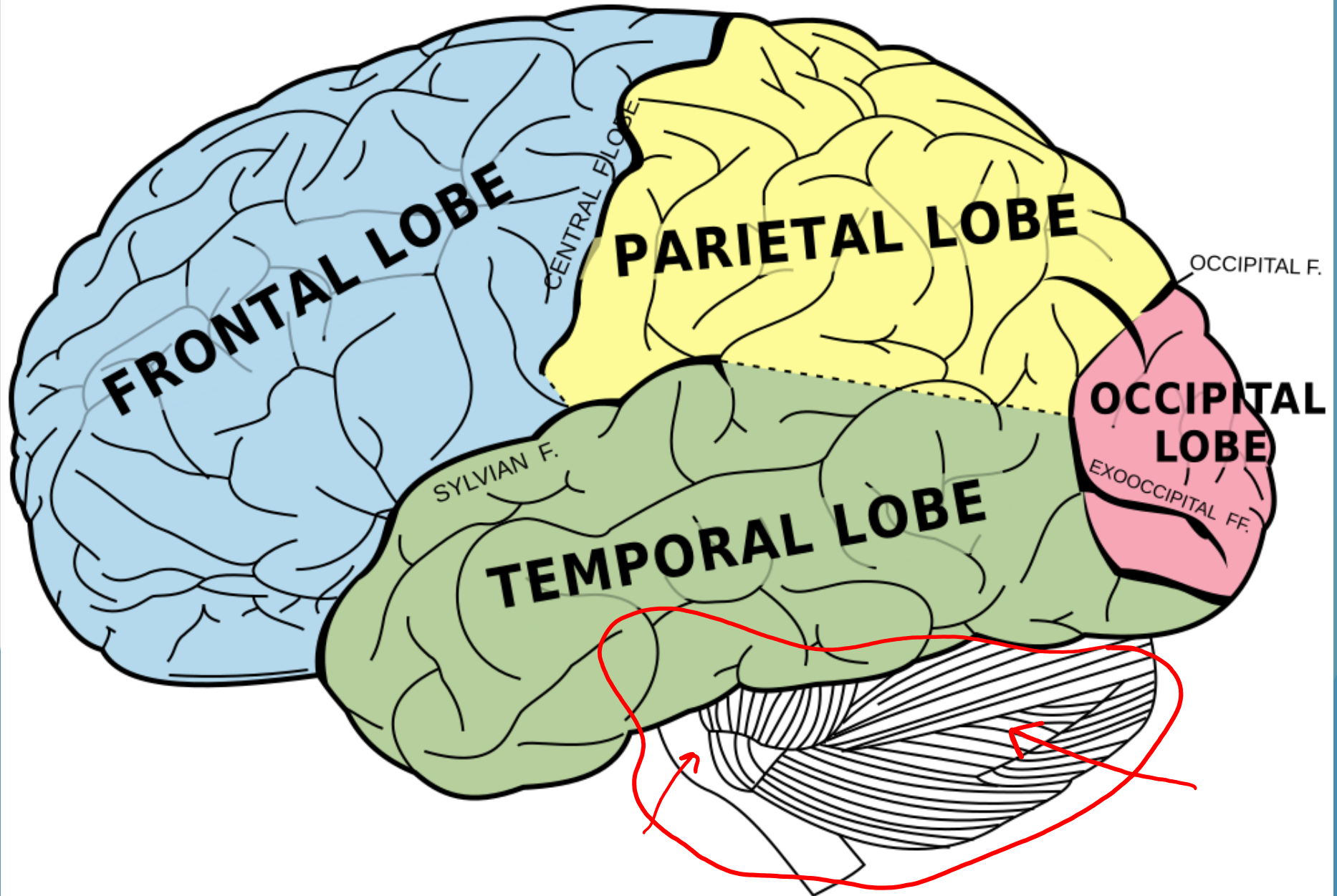
1. Match four structures of the reflex arc numbered in the diagram with their descriptions given below.

Number:	4	1	5	3
Description:	Stimulates a motor neuron	Stimulates a sensory neuron	Stimulates an interneuron	Stimulates an effector

(Record all **four digits** of your answer in the response area at the bottom of the screen.)

Brain: Define each structure

- ▶ Structures: cerebral cortex, hemispheres (left and right) and lobes (4); cerebellum, pons, medulla oblongata, hypothalamus
 - ▶ Cerebrum: thinking part of brain, cerebral cortex is outer wrinkly part (increase surface area)
 - ▶ Left hemisphere - controls right side, math, logic, problem solving
 - ▶ Right hemisphere - controls left side, visual, musical, mental imaging.
 - ▶ 4 lobes
 - ▶ Frontal: fine motor skills, reasoning, higher thinking, speech, planning
 - ▶ Parietal: recognition of stimuli, sensory input
 - ▶ Occipital lobe: visual information
 - ▶ Temporal: auditory information, memory
- ▶ Brainstem: medulla, pons, midbrain (involuntary)

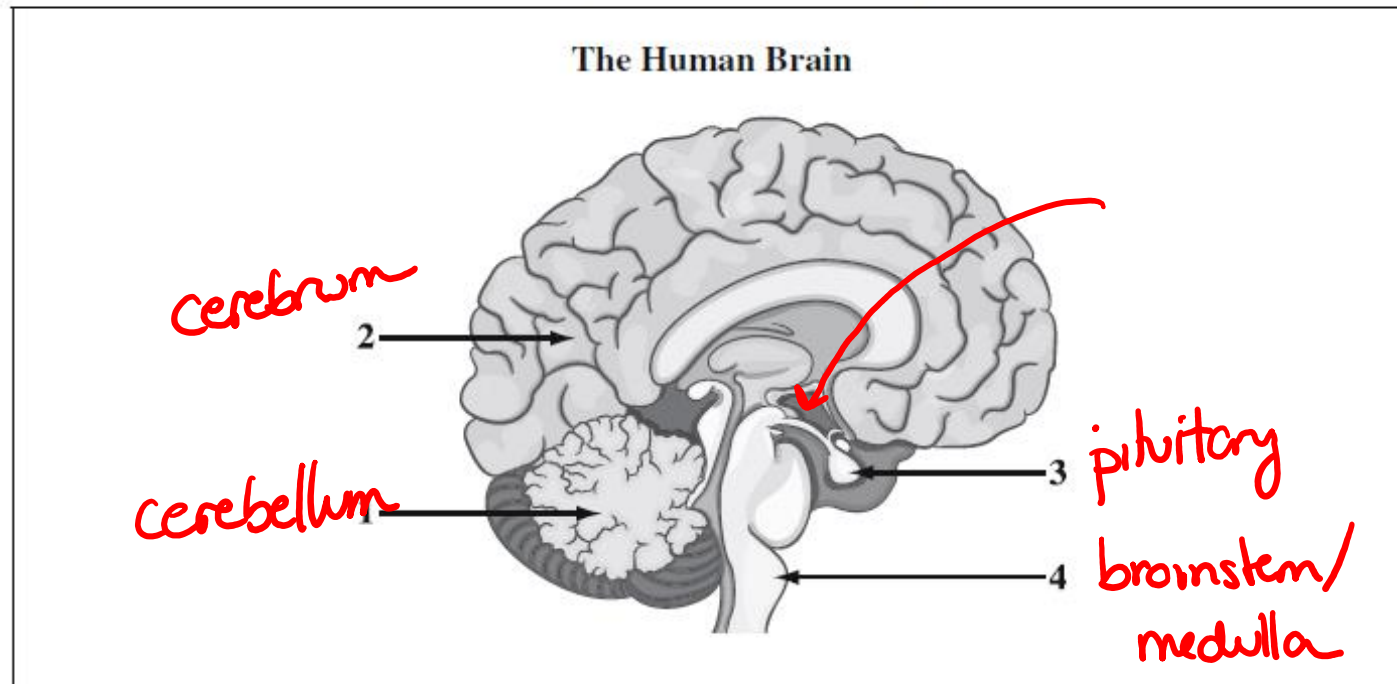


Brain structures

- ▶ Cerebellum - balance and movement
- ▶ Thalamus - relay center for senses
- ▶ Hypothalamus - regulates homeostasis, links nervous and endocrine systems, sends signal to pit gland
- ▶ Pons - sleep, motor control
- ▶ Midbrain - senses, alertness, regulates temperature
- ▶ Medulla oblongata - vital body functions; breathing, heart rate, blood pressure

Practice

Use the following information to answer question 7.



7. The area of the brain that controls the sympathetic and parasympathetic nervous systems is numbered

- A. 1
- B. 2
- C. 3
- D. 4

Practice

Use the following information to answer question 9.

Biofeedback consists of conscious efforts to control body responses that are usually involuntary.

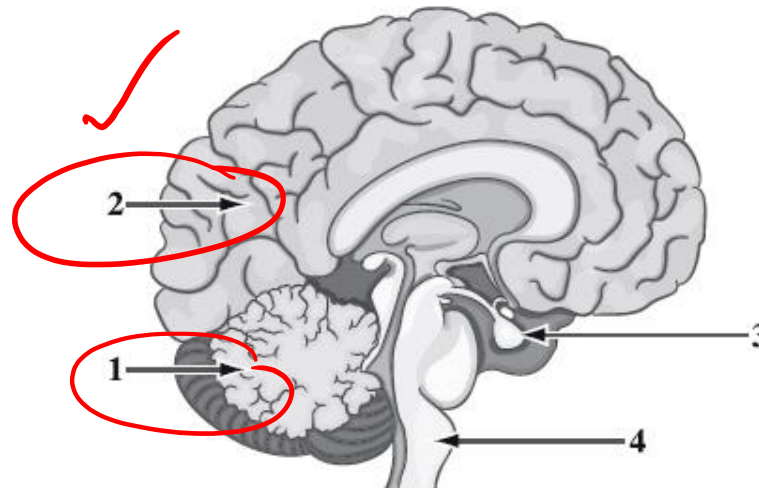
9. The part of the brain in which conscious efforts to control body responses through biofeedback originate is the
- A. medulla
 - ☒ B. cerebrum
 - C. cerebellum
 - D. hypothalamus

Practice

Use the following information to answer questions 10 to 12.

People with Refsum disease cannot metabolize phytanic acid, which results in a buildup of phytanic acid in body tissues and impairs the development of the myelin sheath on neurons. Symptoms of Refsum disease include hearing and vision loss, decreased muscle coordination, and a reduced sense of smell.

The Human Brain



10. In the diagram above, two areas of the brain whose function can be affected in a person with Refsum disease are numbered

- ☒ A. 1 and 2
- ☐ B. 1 and 4
- ☐ C. 2 and 3
- ☐ D. 3 and 4

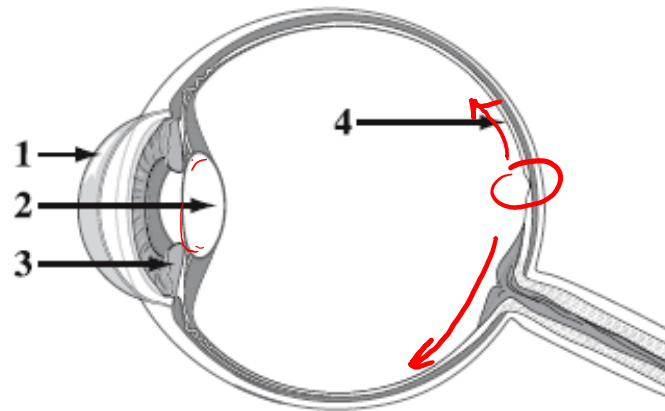
11. Which of the following rows identifies the structure of a neuron affected by a buildup of phytanic acid in people with Refsum disease and describes the expected effect of phytanic acid buildup on nerve impulse transmission?

Row	Neuron Structure	Expected Effect
A.	Axon	Nerve impulses travel faster than normal
B.	Axon	Nerve impulses travel slower than normal
C.	Cell body	Nerve impulses travel faster than normal
D.	Cell body	Nerve impulses travel slower than normal

Use the following additional information to answer question 12.

Symptoms of vision loss in people with Refsum disease include clouding of the lens, a condition known as cataracts, and impaired night vision.

The Human Eye



12. Two structures of the eye that are associated with the symptoms of the type of vision loss in people with Refsum disease are

- A. 1 and 4
- B. 2 and 3
- ☒ C. 2 and 4
- D. 3 and 4

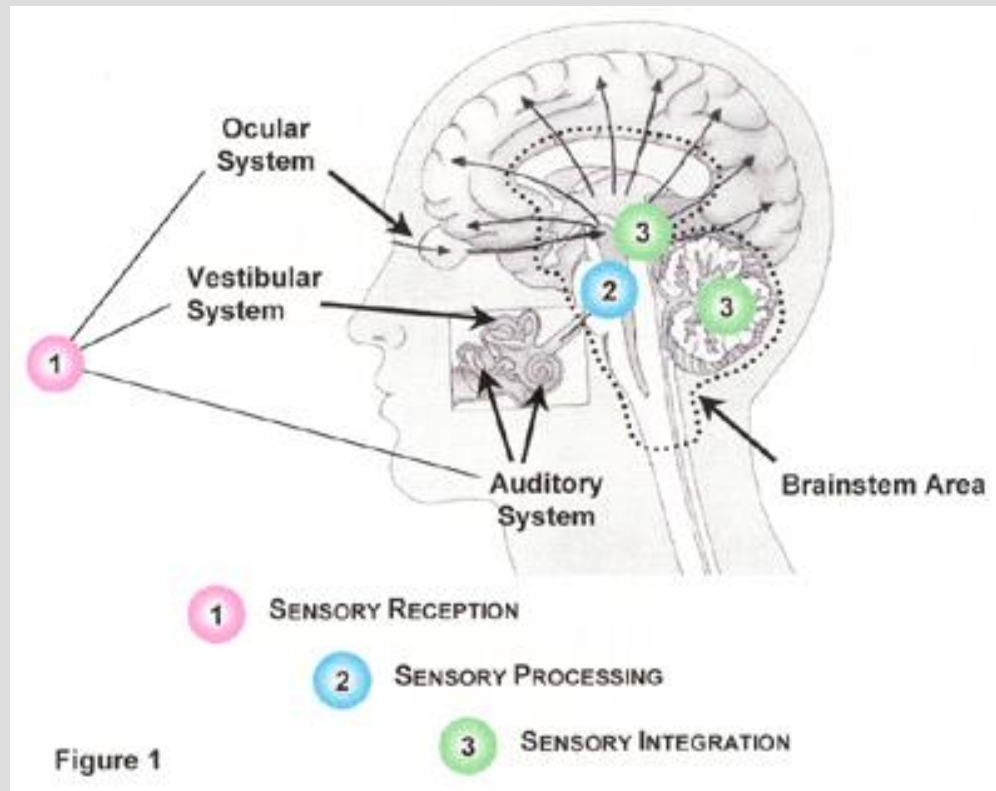
Practice

17. The lobe of the brain in which sounds are **first** analyzed is the

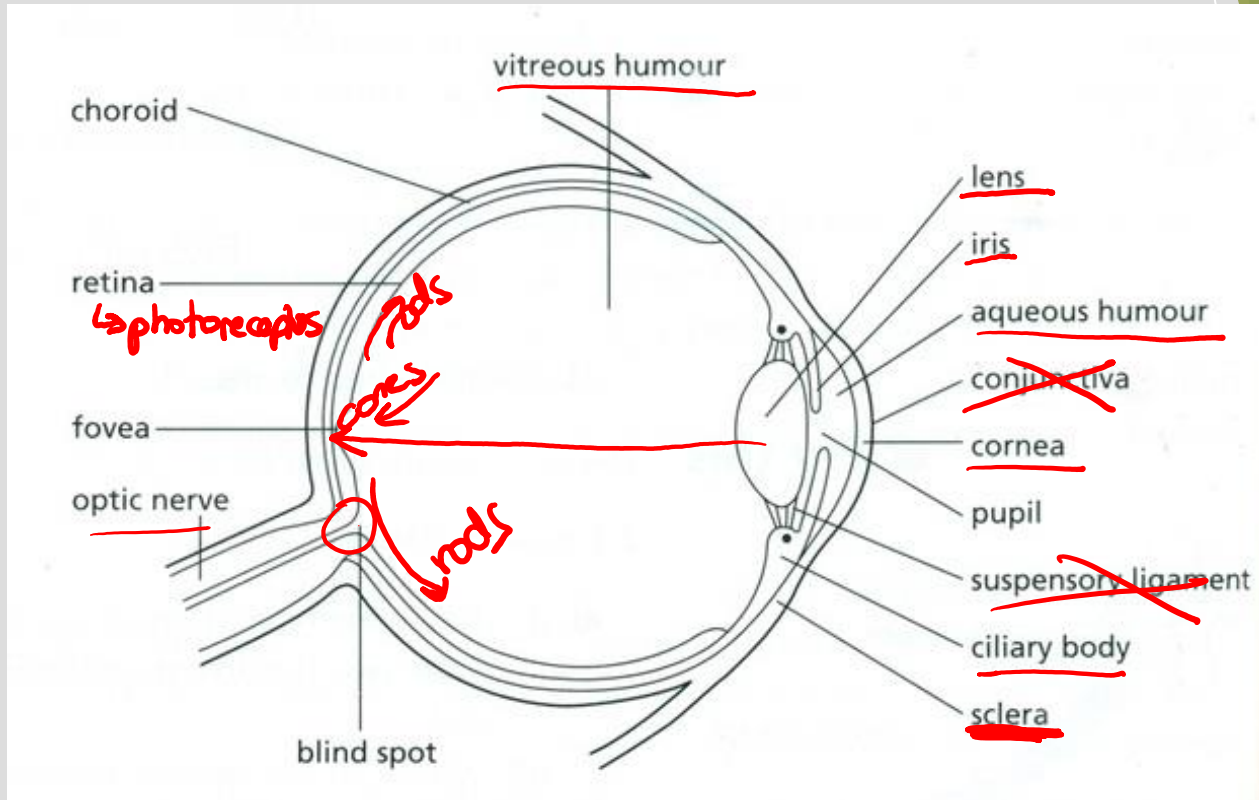
- A. frontal lobe
- B. parietal lobe
- ☒ C. temporal lobe
- D. occipital lobe

Senses

- Sensory reception occurs at the senses. Sensation and perception occur in the brain



Eyes



Eyes

- ▶ Cornea - refracts light towards lens
- ▶ Aqueous humor - fluid between cornea *and the lens*
- ▶ Pupil - opening in iris that allows light to pass in
- ▶ Iris - controls size of pupil
- ▶ Lens - focuses light onto the retina
- ▶ Vitreous humor - fluid in eye, gives it shape
- ▶ Sclera - outer part of eye, provides structure support and protection
- ▶ Retina - senses light and sends signal through optic nerve to brain
 - ▶ Contains rods that detect dark/black and white
 - ▶ Contains cones that detect colour, detail, lots of cones near the fovea centralis
- ▶ Choroid - nourish the eye, absorb light

Practice

Use the following information to answer numerical-response question 1.

The pupillary reflex is a reflex that occurs in response to varying intensities of light entering the eye.

Some Events in the Pupillary Reflex

- 1 Pupil diameter changes
- 2 Motor neuron depolarizes
- 3 Light receptors stimulated
- 4 Sensory neuron depolarizes

Numerical Response

1. The order in which the events numbered above occur during the pupillary reflex is 3, 4, 2, and 1.

(Record all **four** digits of your answer in the numerical-response section on the answer sheet.)

Use the following information to answer numerical-response question 3.

Some Structures of the Nervous System

- 1 Optic nerve
- 2 Photoreceptor
- 3 Occipital lobe

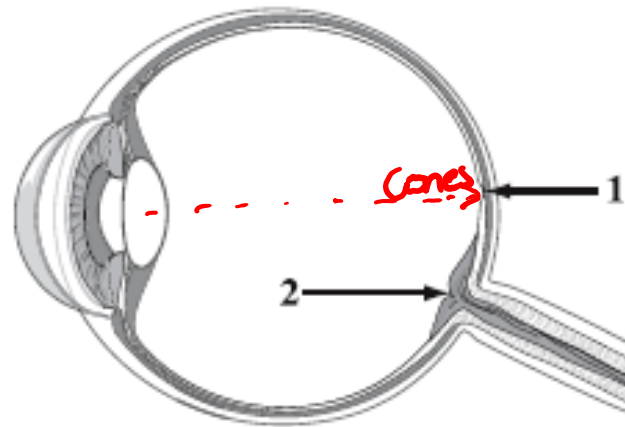
Numerical Response

3. After light enters the eye, the order in which the structures numbered above are stimulated is 2, 1, and 3.

Use the following information to answer question 15.

Erectile dysfunction, which is defined as the inability to maintain an erection, can sometimes be treated with the drug sildenafil citrate. A side effect that may occur while taking sildenafil citrate is difficulty distinguishing between the colours blue and green.

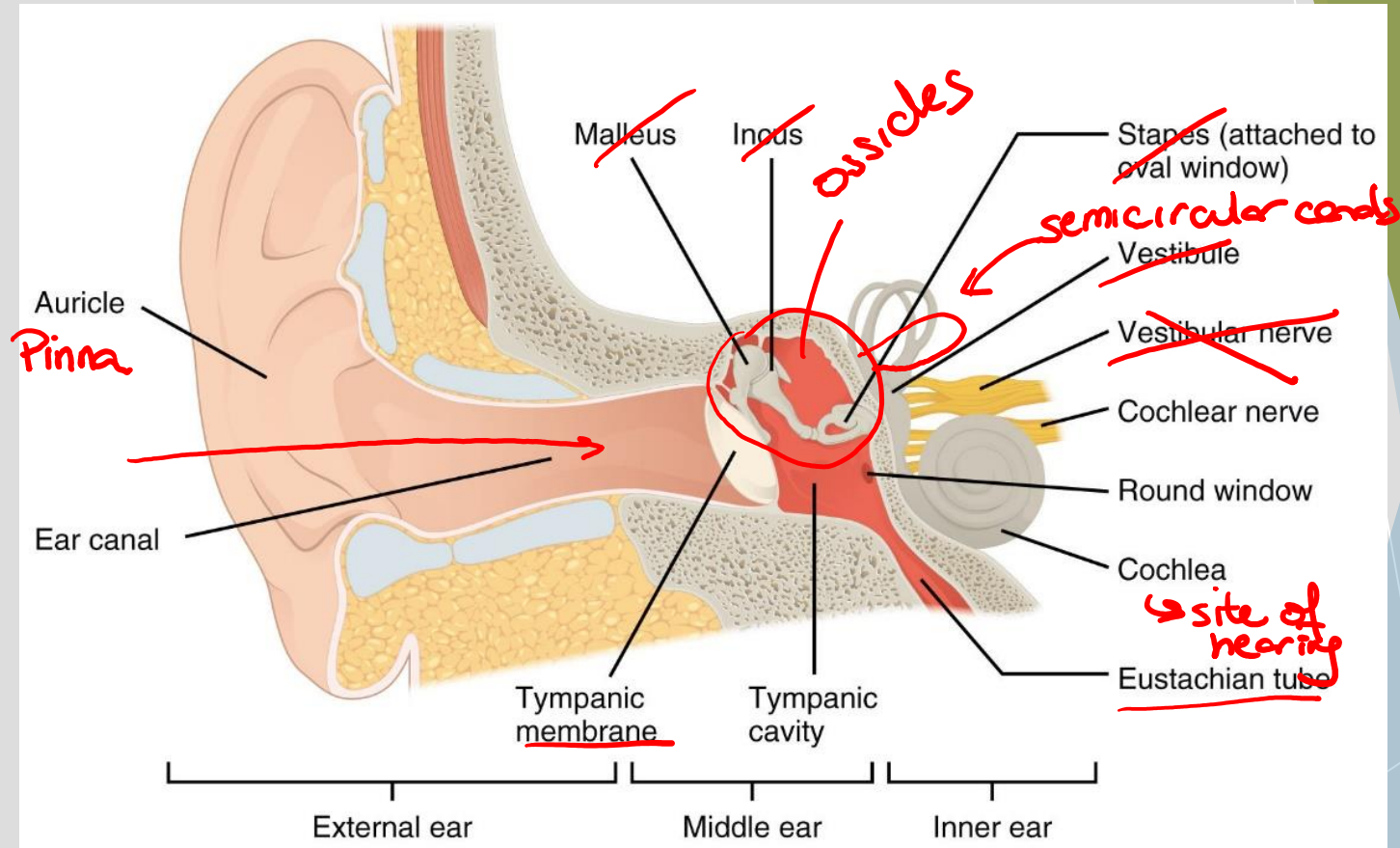
The Human Eye



15. The cells in the eye that are affected by sildenafil citrate are primarily found in

- A. location 1, and these cells are known as rods
- ☒ B. location 1, and these cells are known as cones
- C. location 2, and these cells are known as rods
- D. location 2, and these cells are known as cones

Ear



Structures

- ▶ Pinna - outer part of the ear, some mammals can control
- ▶ Auditory canal - tube that carries sound to ear drum
- ▶ Tympanic membrane - ear drum, vibrates in response to sound waves
- ▶ Ossicles (3) - smallest bones in body, transmit and amplify vibration from ear drum to fluid filled cochlea through oval window
- ▶ Organ of Corti - inside the cochlea, has hair cells in fluid that move in response to vibrations from ossicles, send nerve impulse *→ auditory receptor*
- ▶ Semicircular canals - fluid filled canals that send signals concerning balance and head position
- ▶ Eustachian tube - leads to throat, controls pressure inside the ear (ear popping)

Practice

Use the following information to answer numerical-response question 4.

When a car's airbag deploys, there is a rapid increase in air pressure that results from the rapid expansion of gases inside the airbag. This expansion causes pressure waves that can damage the inner ear.

Structures of the Human Ear

- 1 Ossicles
- 2 Cochlea
- 3 Auditory canal
- 4 Tympanic membrane

Numerical Response

4. The sequence in which the highly compressed pressure waves created by the deployment of an airbag travel through the structures of the human ear is _____, _____, _____, and _____.

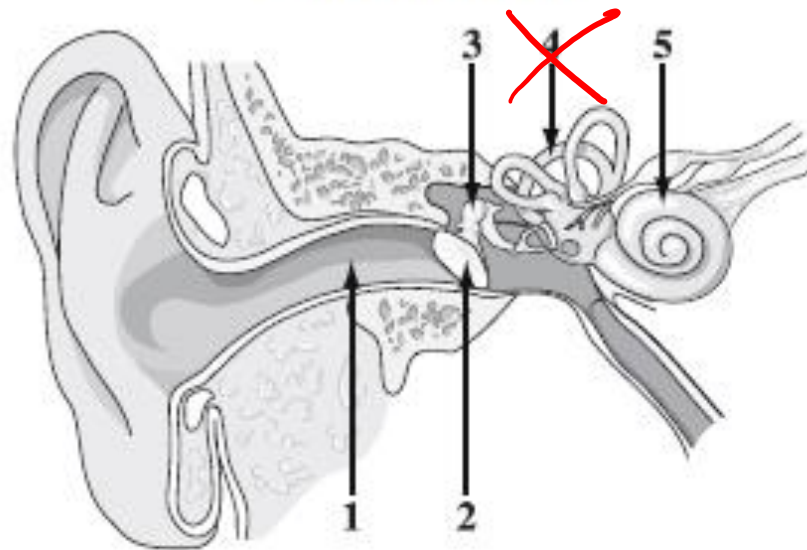
3 4 1 2
3412

16. Which of the following rows identifies the structure of the ear that converts mechanical vibrations into electrochemical impulses and the structure of the ear where electrochemical impulses are first produced?

Row	Structure That Converts Vibrations into Impulses	Structure Where <u>Impulses</u> are First Produced
A.	Cochlea	Ossicles
B.	Cochlea	Organ of Corti
C.	Semicircular canals	Ossicles
D.	Semicircular canals	Organ of Corti

Use the following information to answer numerical-response question 5.

The Human Ear

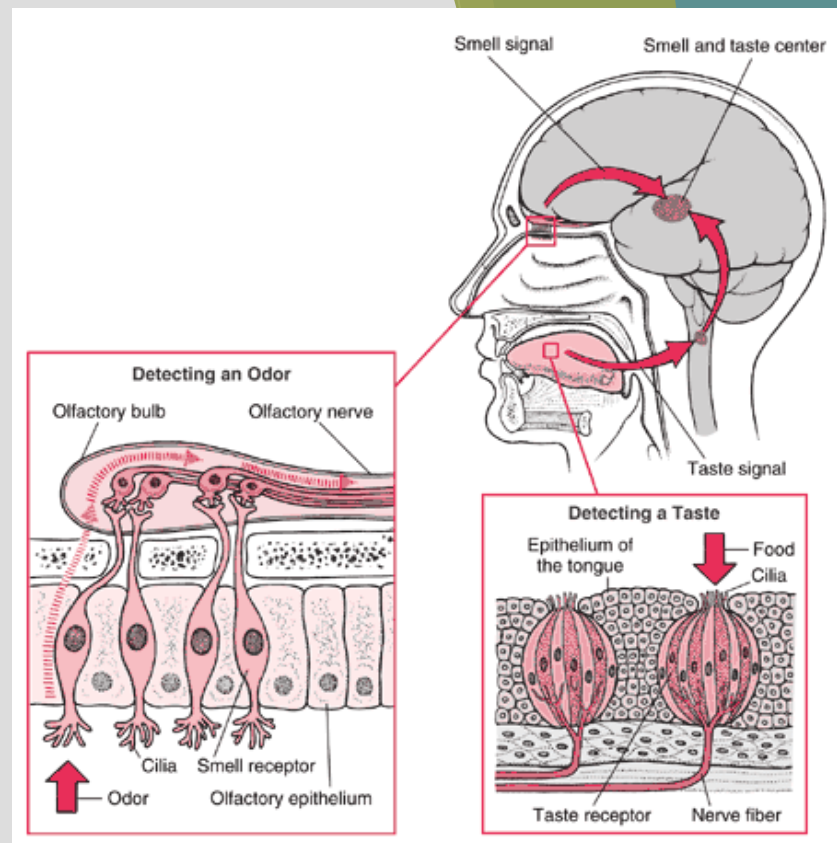
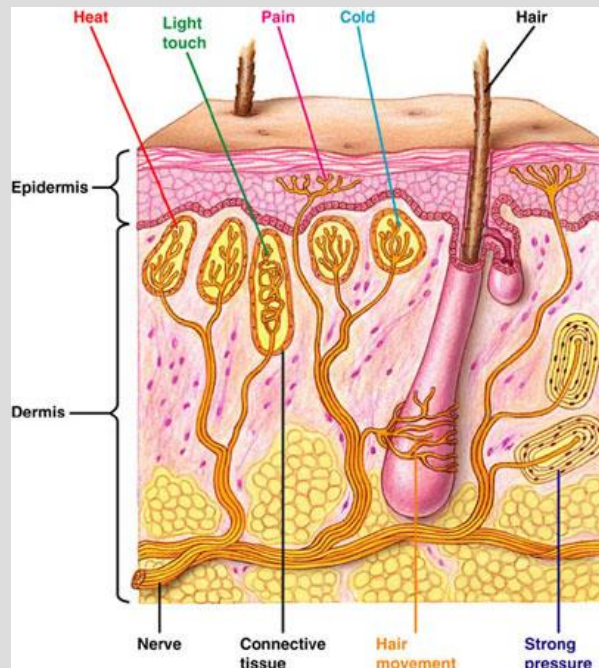


Numerical Response

5. Four structures numbered above that are involved in hearing are 1, 2, 3, and 5.

Other senses

- ▶ Tastes are detected by chemoreceptors in the taste buds of the tongue. Smells are detected by chemoreceptors in the nose.
- ▶ The skin contains receptors for light touch, pressure, pain, heat, and cold.



Practice

~~*~~
not in notes

Use the following information to answer question 3.

In a research study on the detection of odours, a group composed of men, women of reproductive age, postmenopausal women, and children were asked to smell gradually decreasing concentrations of specific familiar chemicals. The women of reproductive age were more able to detect weaker odours than were the men, postmenopausal women, and the children. The researchers concluded that female sex hormones might increase sensitivity to familiar odours.

3. The inference that can be made from the study described above is that compared to men, postmenopausal women, and children, women of reproductive age have
- ~~A.~~ more receptors for odour detection
 - ☒ B. a lower threshold level for familiar odours
 - ~~C.~~ the ability to interpret odours more quickly
 - ~~D.~~ the ability to adapt to familiar odours more quickly

Use the following information to answer numerical-response question 6.

Some Cell Types

- 1 Rod cells
- 2 Cone cells
- 3 Cochlea hair cells
- 4 Taste receptor cells
- 5 Semi-circular canal hair cells

Some Disorders

Colour-blindness—a genetic disorder that results in the inability to detect certain colours of light

Neural deafness—the inability to detect sound as a result of damage to sensory structures in the inner ear

Permanent vertigo—a severe balance disorder that usually results from physical trauma to the ear

Numerical Response

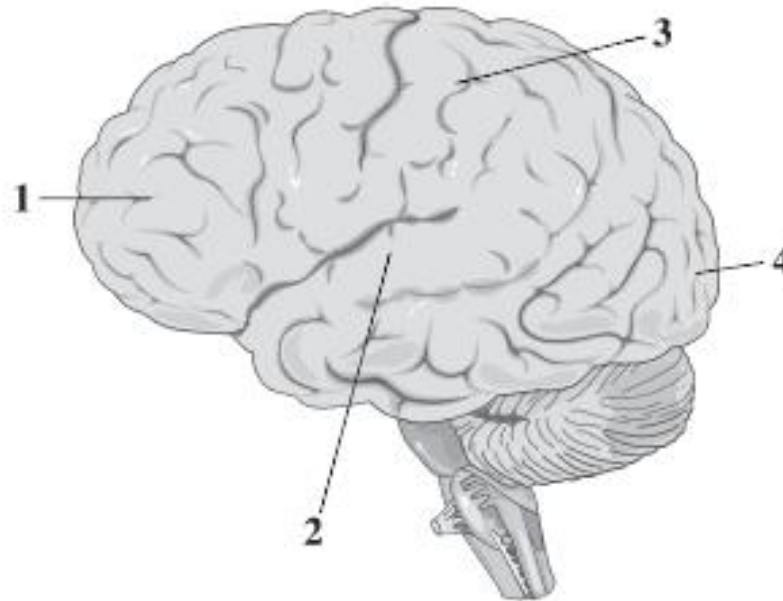
6. Match three of the cell types numbered above with the disorder that is most closely associated with each cell type.

Number:	<u>2</u>	<u>3</u>	<u>5</u>
Disorder:	Colour-blindness	Neural deafness	Permanent vertigo

Use the following information to answer question 5.

Absinthe, a green liqueur made from the wormwood plant, can cause both auditory and visual hallucinations, along with other effects, when consumed.

The Human Brain



5. If a person experienced both auditory and visual hallucinations after consuming absinthe, the regions of the brain numbered in the diagram that would be affected are

A. 1 and 2

B. 1 and 4

C. 2 and 3

D. 2 and 4

Unit A - Outcome 2

- ▶ identify the principal endocrine glands of humans; i.e., the hypothalamus/pituitary complex, thyroid, parathyroid, adrenal glands and islet cells of the pancreas
- ▶ describe the function of the hormones of the principal endocrine glands, i.e., thyroid-stimulating hormone (TSH)/thyroxine, calcitonin/parathyroid hormone (PTH), adrenocorticotrophic hormone (ACTH)/cortisol, glucagon/insulin, human growth hormone (hGH), antidiuretic hormone (ADH), epinephrine, aldosterone, and describe how they maintain homeostasis through feedback
- ▶ explain the metabolic roles hormones may play in homeostasis; i.e., thyroxine in metabolism; insulin, glucagon and cortisol in blood sugar regulation; hGH in growth; ADH in water regulation; aldosterone in sodium ion regulation
- ▶ explain how the endocrine system allows humans to sense their internal environment and respond appropriately; *e.g., calcium balance, osmotic pressure of blood*
- ▶ compare the endocrine and nervous control systems and explain how they act together; *e.g., stress and the adrenal gland*
- ▶ describe, using an example, the physiological consequences of hormone imbalances; i.e., diabetes mellitus

Hormone Regulation - Negative feedback

- ▶ When a hormone's level rise, the organ that releases that hormone is turned off

- ▶ Thermostat analogy

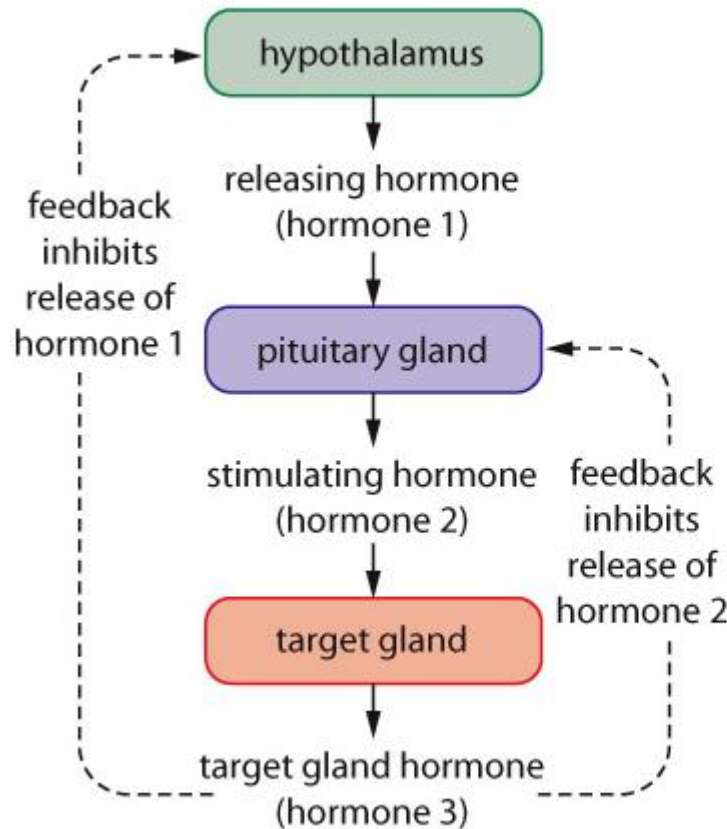
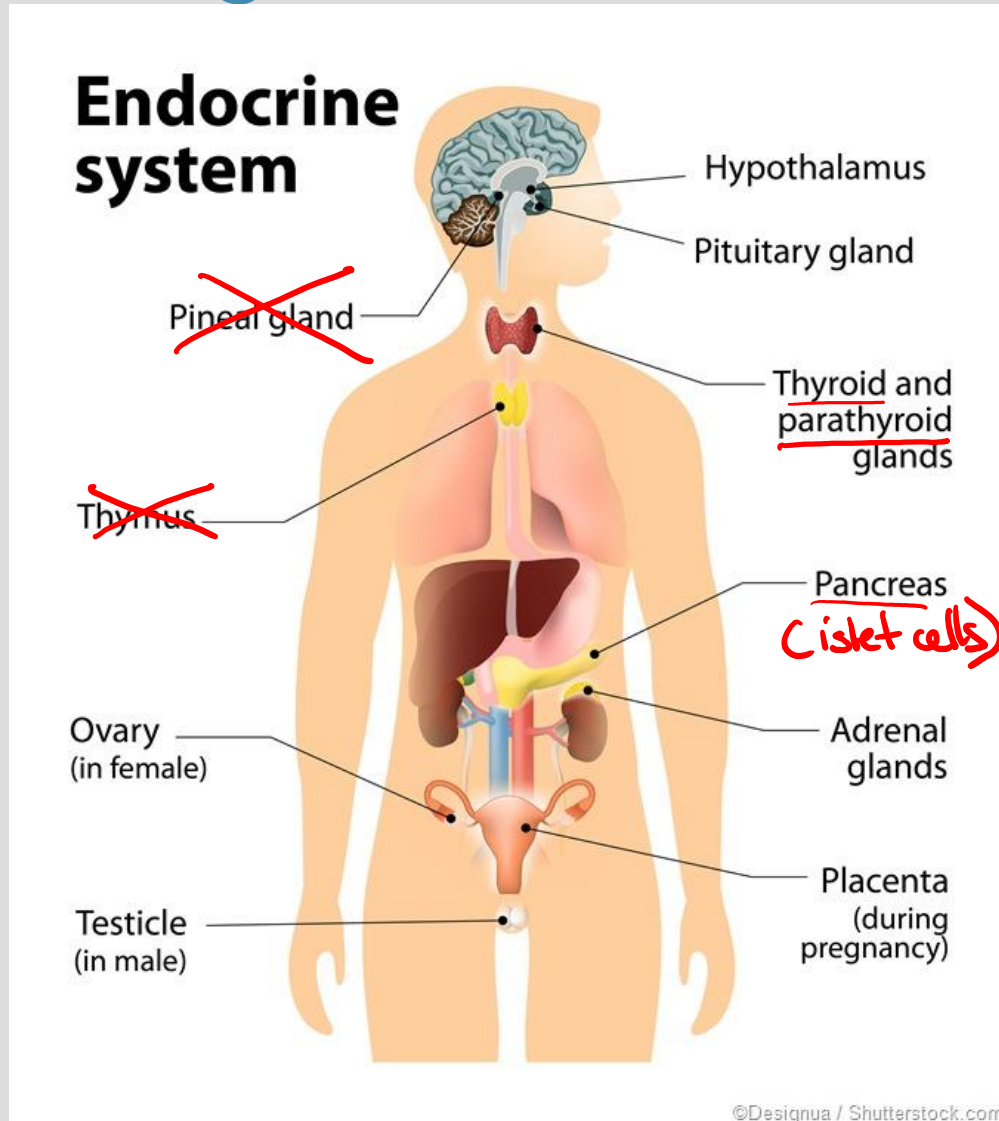


Figure 13.10 The generalized regulatory pathway of tropic hormones. The target gland hormone will affect other tissues in the body, such as the bones and muscles. How are tropic hormones regulated by negative feedback?

Endocrine glands



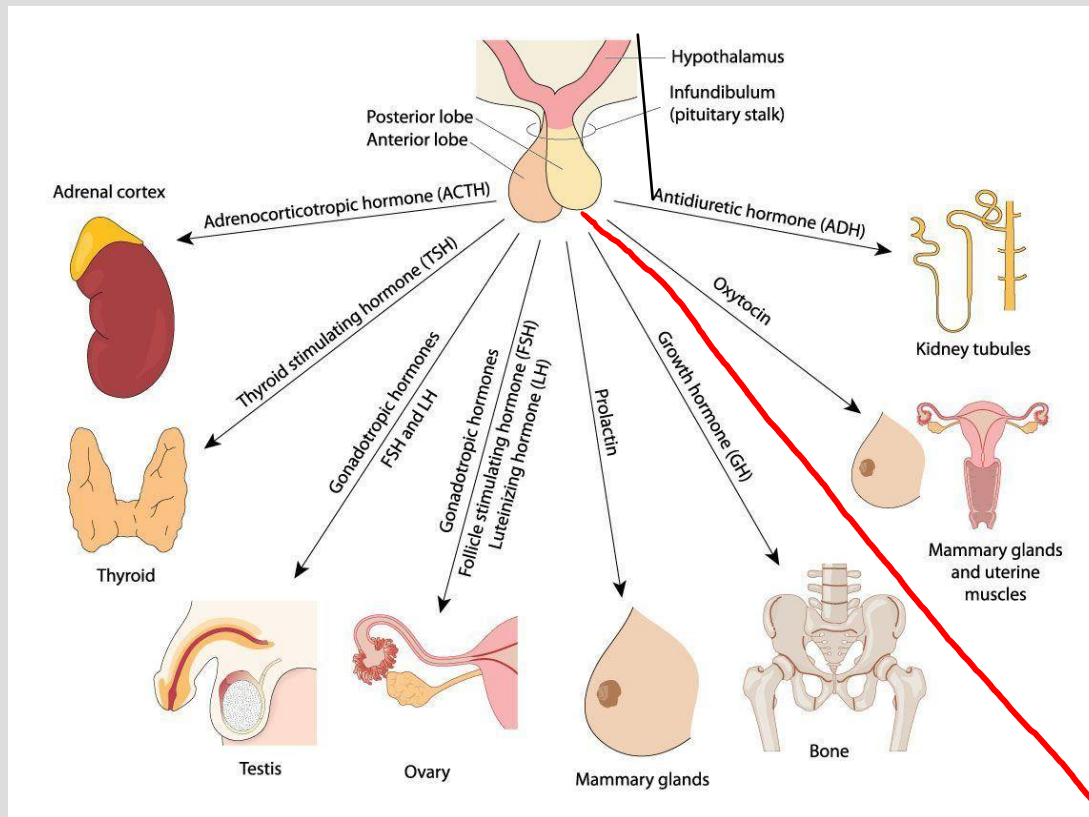
Endocrine glands

- ▶ Hypothalamus, pituitary, thyroid, parathyroid, adrenal glands and islet cells of the pancreas
- ▶ Hypothalamus links the nervous and endocrine system
- ▶ Endocrine system is slower and more general than the nervous system (hormones release into blood)

Hormones

- ▶ Hormone secretion is regulated by the nervous system, other hormones, or negative feedback mechanisms.
- ▶ The hypothalamus regulates the pituitary gland, which secretes tropic hormones that affect various endocrine glands.
- ▶ Thyroid-stimulating hormone (TSH)/thyroxine, calcitonin/parathyroid hormone (PTH), adrenocorticotrophic hormone (ACTH)/cortisol, glucagon/insulin, human growth hormone (hGH), antidiuretic hormone (ADH), epinephrine, aldosterone, *LH/FSH, testosterone* 
- ▶ Pituitary Hormones
 - ▶ Anterior: TSH, ACTH, FSH/LH, hGH, prolactin, endorphins
 - ▶ Posterior: Oxytocin, ADH

Pituitary Gland - Master Gland





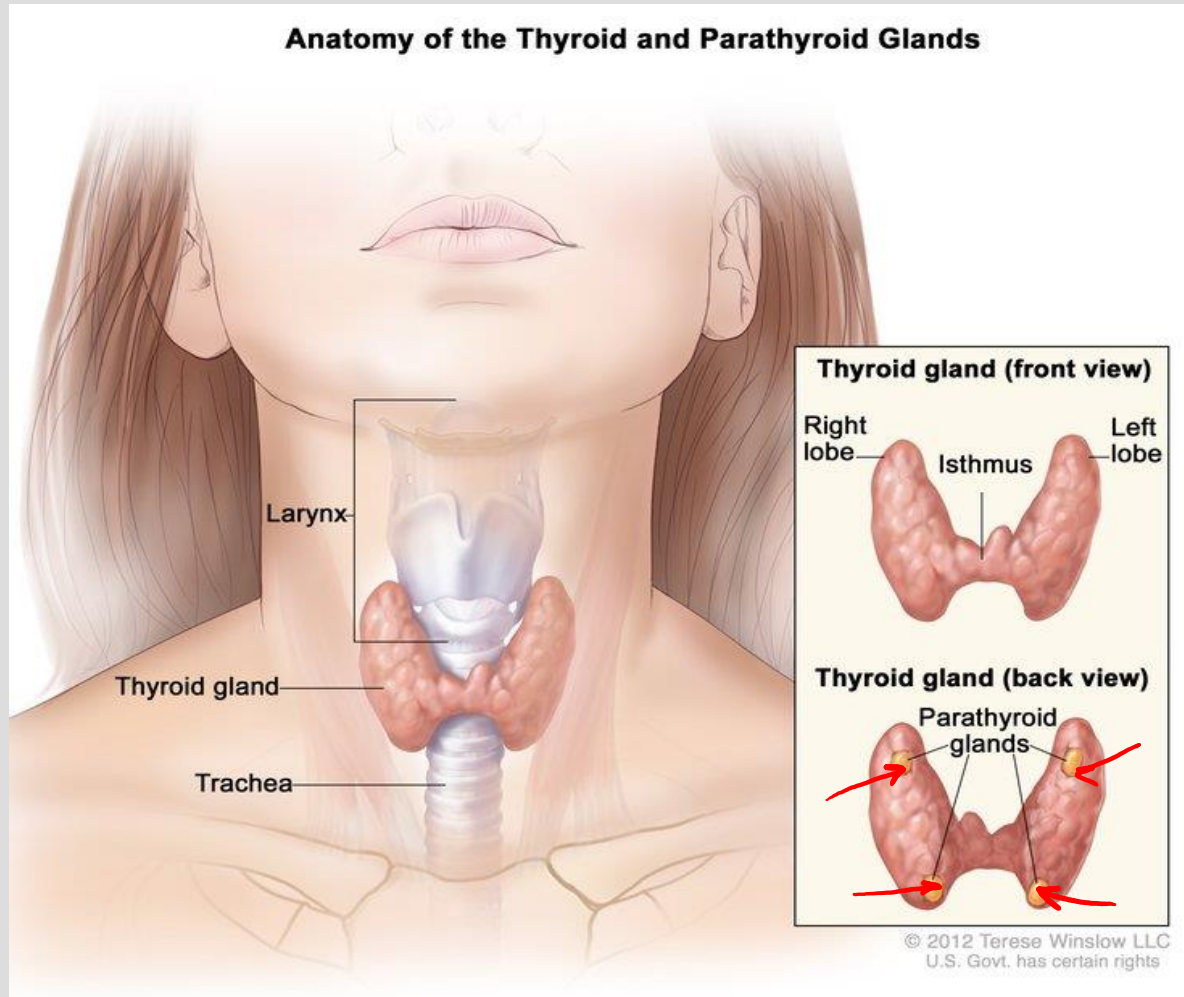
Human Growth Hormone

- ▶ Human Growth Hormone:
 - ▶ Mainly affects bone and muscle growth, increased during puberty.
 - ▶ Brain storm: What happens with too much or too little?

too much → gigantism

too little → dwarfism.

Thyroid



Thyroid

► Roles:

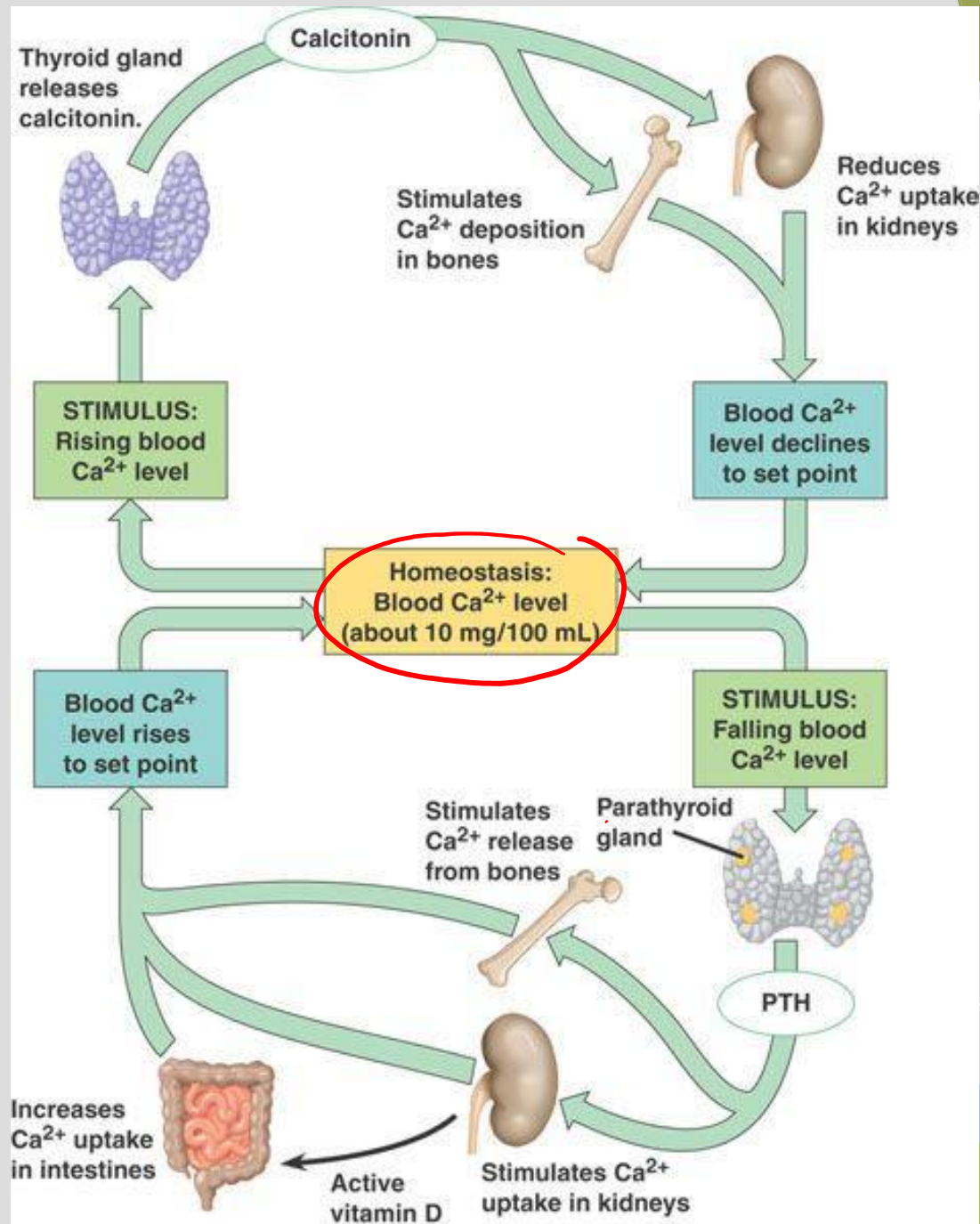
► Metabolism

- Stimulated by TSH (from ant. pit.)
- Releases T3, T4 (thyroxine), both increase metabolism
 - Use up blood sugar, burn fat, increase temp

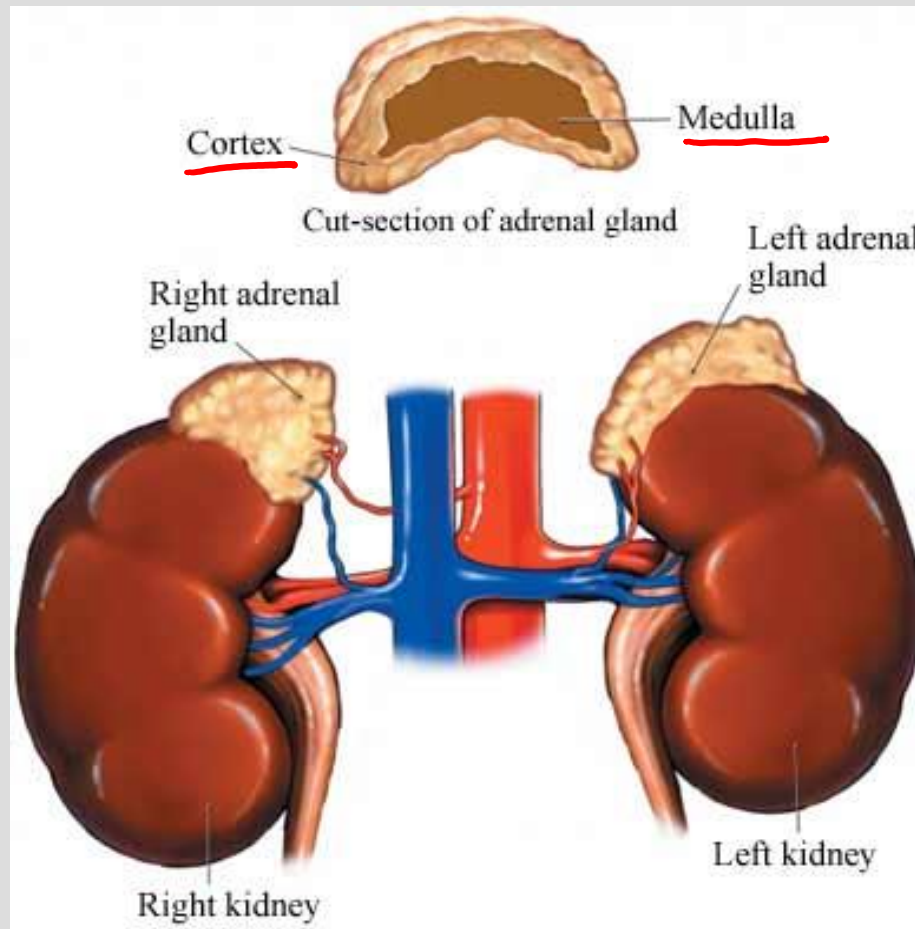
► Calcium levels

- Releases calcitonin (when blood Ca^{2+} levels high)
↳ 'calci-bone-in'
- Parathyroid releases PTH when blood calcium levels are low

antagonistic hormones [



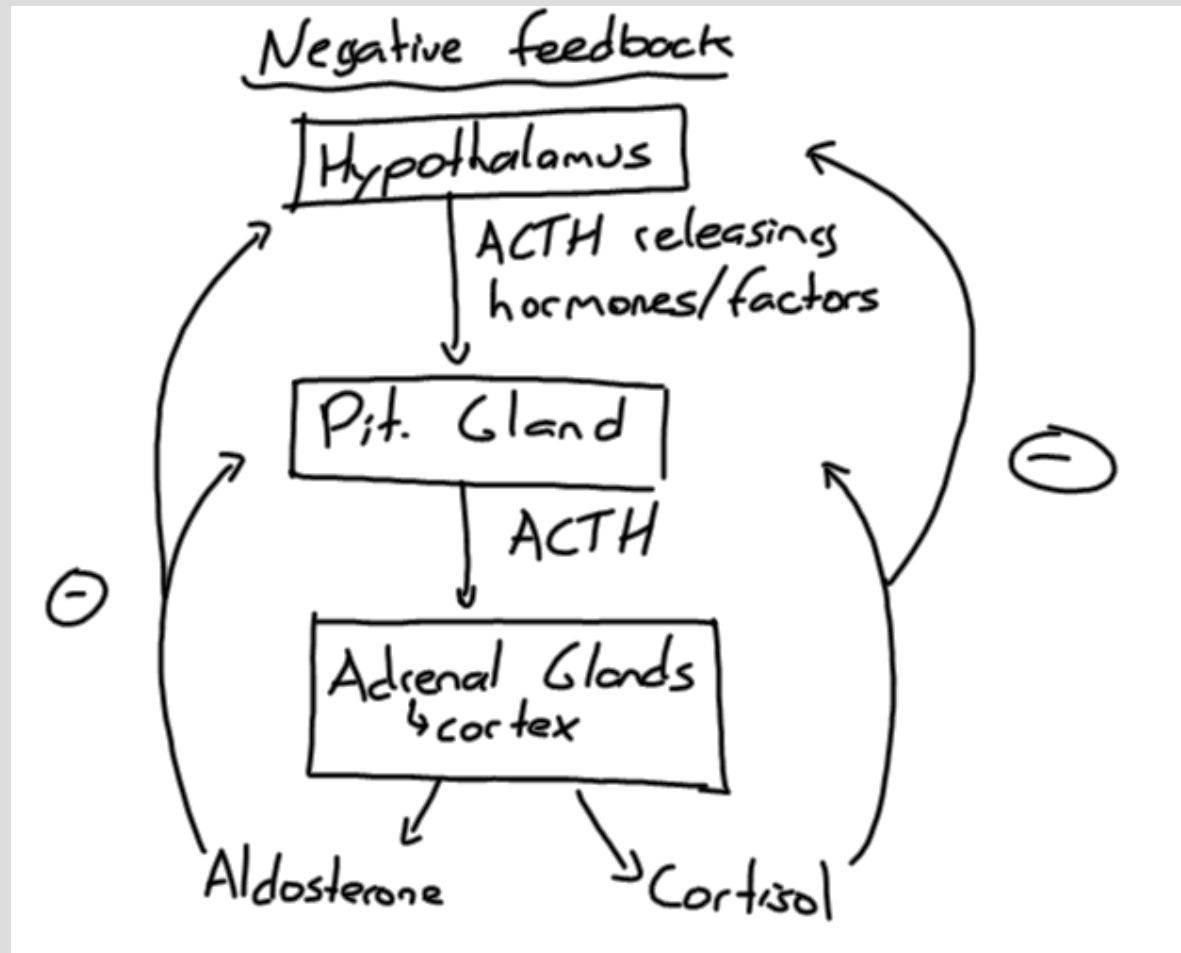
Adrenal Glands



<u>Cortex</u>	<u>Medulla</u>
- stimulated by the endocrine system ↳ <u>ACTH</u> (pit. gland) - <u>long-term</u> stress response - releases: ↳ aldosterone - ↑ blood pressure by reabsorbing H_2O - by ↑ Na^+ reabsorption ↳ <u>Cortisol</u> - reduces pain + <u>inflames</u> - ↑ energy available	- stimulated by the nervous system ↳ <u>Sympathetic</u> - <u>short-term</u> - releases: ↳ <u>epinephrine</u> + <u>norepinephrine</u> - ↳ turns on 'fight or flight' response - increases B.P. and H.R. - dilates bronchioles - decrease digestion

- breakdown of protein and converting amino acid to glucose

Negative Feedback loop



Practice

* know hypo — thyroxine
hyper —

Use the following information to answer the next question.

Pituitary dwarfism is an inherited condition that occurs in some dog breeds. A dog with pituitary dwarfism frequently has decreased secretion of several pituitary hormones, which results in abnormal functioning of other endocrine glands.

↓ TSH

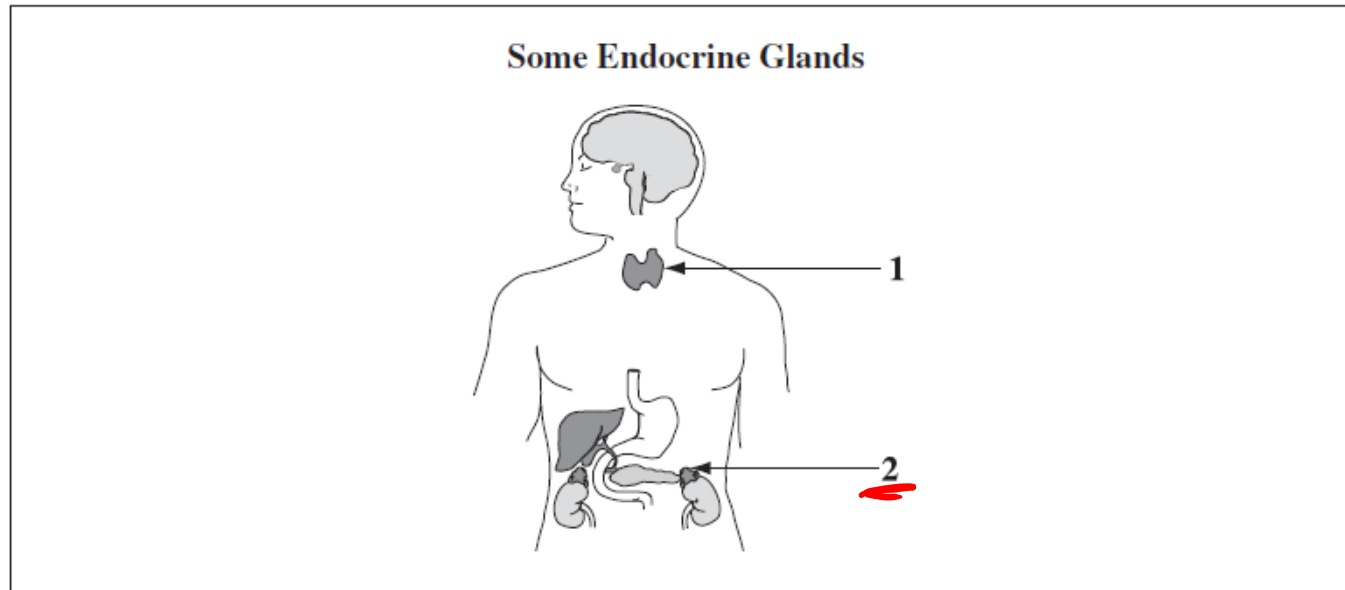
—based on *Canine Inherited Disorders Database*, 2010

Canine Inherited Disorders Database. 2010. Inherited endocrine disorders: Pituitary dwarfism (Hypopituitarism). *Canine Inherited Disorders Database*. <http://www.upei.ca/cidd/intro.htm>.

7. Two expected symptoms of pituitary dwarfism in dogs are

- ☒ A. hypothyroidism and decreased metabolic rate
- ☐ B. hyperthyroidism and increased metabolic rate
- ☐ C. hypoparathyroidism and decreased metabolic rate
- ☐ D. hyperparathyroidism and increased metabolic rate

Use the following diagram to answer the next question.



12. In the diagram above, the gland that releases a hormone in response to short-term stress is numbered

- A. 1, and it releases the hormone after being stimulated by the nervous system
- ☒ B. 2, and it releases the hormone after being stimulated by the nervous system
- C. 1, and it releases the hormone after being stimulated by the endocrine system
- D. 2, and it releases the hormone after being stimulated by the endocrine system

Use the following information to answer the next two questions.

A man had high blood pressure and experienced muscle weakness and paralysis. These symptoms were caused by long-term exposure to licorice root, which the man consumed in his daily tea. Licorice root contains a substance that mimics the effects of high amounts of aldosterone and cortisol.

—based on Lin et al., 2003

Lin, Shih-Hua, Sung-Sen Yang, Tom Chau, and Mitchell L. Halperin. 2003. An unusual cause of hypokalemic paralysis: Chronic licorice ingestion. *American Journal of the Medical Sciences* 325 (3): 153–156.

10. The man's symptoms were caused by a substance that mimics the effects of hormones secreted by the

- A. thyroid gland
- B. hypothalamus
- ☒ C. adrenal glands
- D. pituitary gland

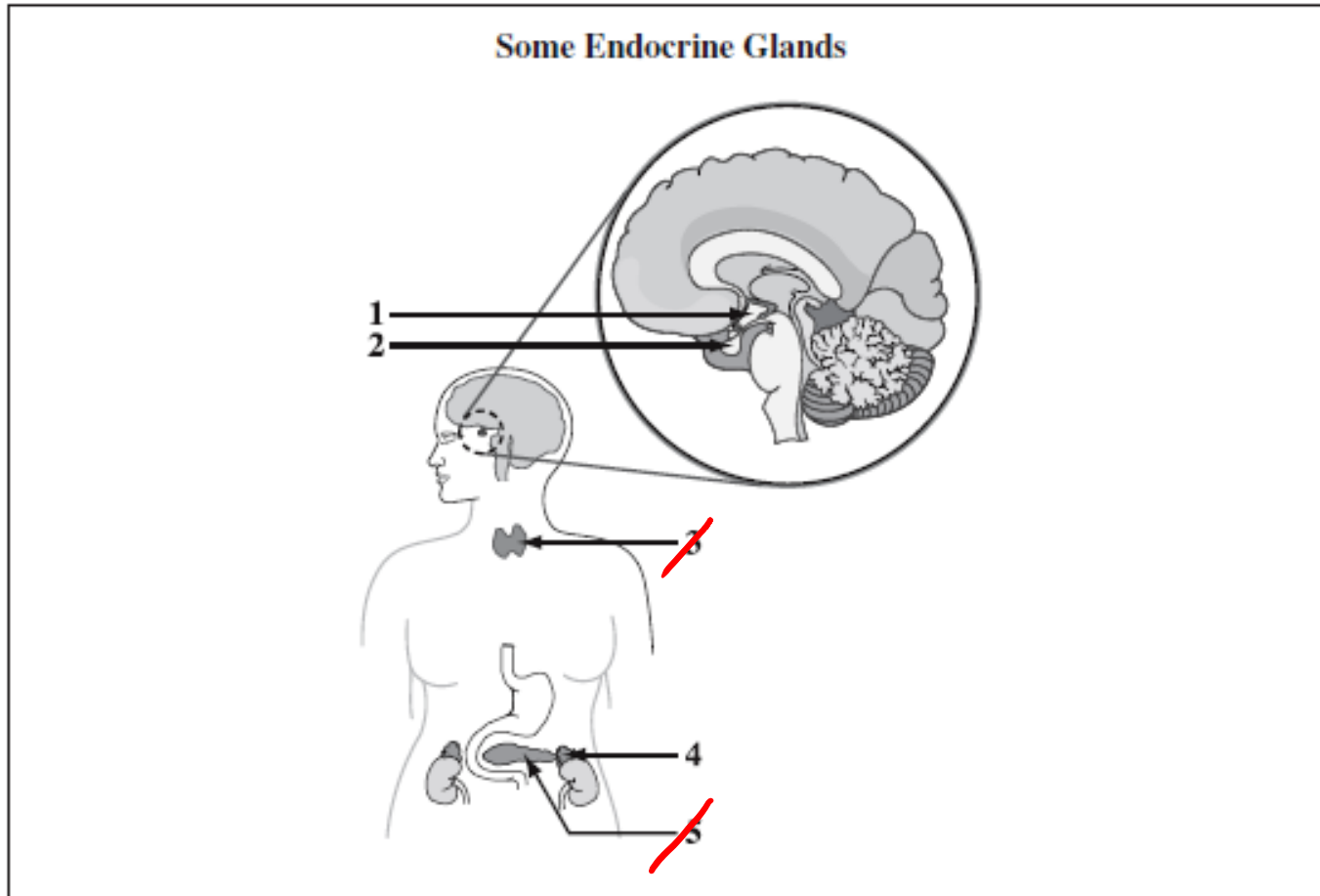
11. A higher-than-normal amount of aldosterone causes

- ☒ A. increased sodium reabsorption by the kidneys
- B. decreased sodium reabsorption by the kidneys
- C. decreased water reabsorption by the kidneys
- D. increased glucose reabsorption by the kidneys

20. Which of the following rows identifies the hormone that stimulates the release of cortisol, the gland that secretes cortisol, and an effect of cortisol?

Row	Hormone	Gland	Effect
A.	ADH	Adrenal gland	Increased conversion of amino acids to glucose
B.	ADH	Pituitary gland	Increased water reabsorption
C.	ACTH	Adrenal gland	Increased conversion of amino acids to glucose
D.	ACTH	Pituitary gland	Increased water reabsorption

Use the following information to answer numerical-response question 7.



Numerical Response

7. The three glands numbered above that are involved in the feedback control of cortisol secretion are 1, 2, and 4.

Use the following information to answer questions 21 and 22.

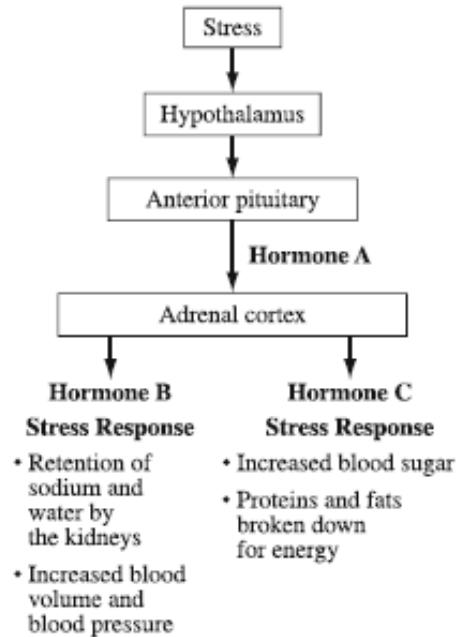
Uncontrolled cell division in cells of the thyroid can lead to the formation of a tumour in the thyroid gland. Thyroid tumours are often treated with a combination of surgery, radioactive iodine, and thyroid medication.

21. A change in hormone level that would be expected immediately following surgical removal of the thyroid gland is
- A. a decrease in both thyroxine and TSH levels
 - B. an increase in both thyroxine and TSH levels
 - C. an increase in thyroxine level and a decrease in TSH level
 - ☒ D. a decrease in thyroxine level and an increase in TSH level
22. Following surgical removal of the thyroid gland, thyroid medication is often prescribed in order to increase
- A. sodium ion reabsorption and increase water retention
 - B. sodium ion reabsorption and decrease water retention
 - ☒ C. the rate of metabolism and increase the rate of heat production
 - D. the rate of metabolism and decrease the rate of heat production

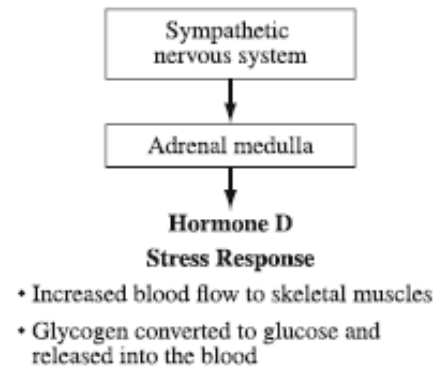
Use the following information to answer numerical-response question 8 and question 24.

Stress Response Flowchart

Long-term response to stress



Short-term response to stress



Some Hormones

- 1 ACTH
- 2 Cortisol
- 3 Aldosterone
- 4 Epinephrine

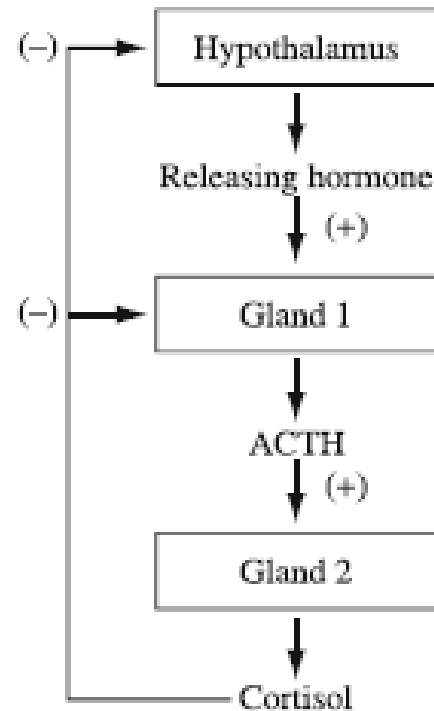
Numerical Response

8. Match each of the hormones numbered above with the hormones represented in the stress response flowchart below.

Number: _____
Hormone: A B C D

24. The short-term response to stress occurs faster than the long-term response to stress because
- A. blood from the adrenal gland travels faster than does blood from the pituitary gland
 - ☒ B. the adrenal gland responds to nervous stimulation, which is faster than hormonal stimulation
 - C. hormones from the adrenal gland act on cells more quickly than hormones from the pituitary gland
 - D. the adrenal gland responds to hormonal stimulation, which is faster than nervous stimulation

The hormone ACTH helps to regulate the nervous system. It also helps the body respond to changes in sleep patterns. The release of ACTH is suppressed during sleep but increases before a person awakes. The regulation of ACTH is shown in the feedback loop below.



26. A possible reason for the suppression of ACTH during sleep is that during sleep, the
- A. secretion of cortisol is decreased
 - B. activity in the hypothalamus is increased
 - C. activity in the pituitary gland is increased
 - D. secretion of releasing hormone is decreased

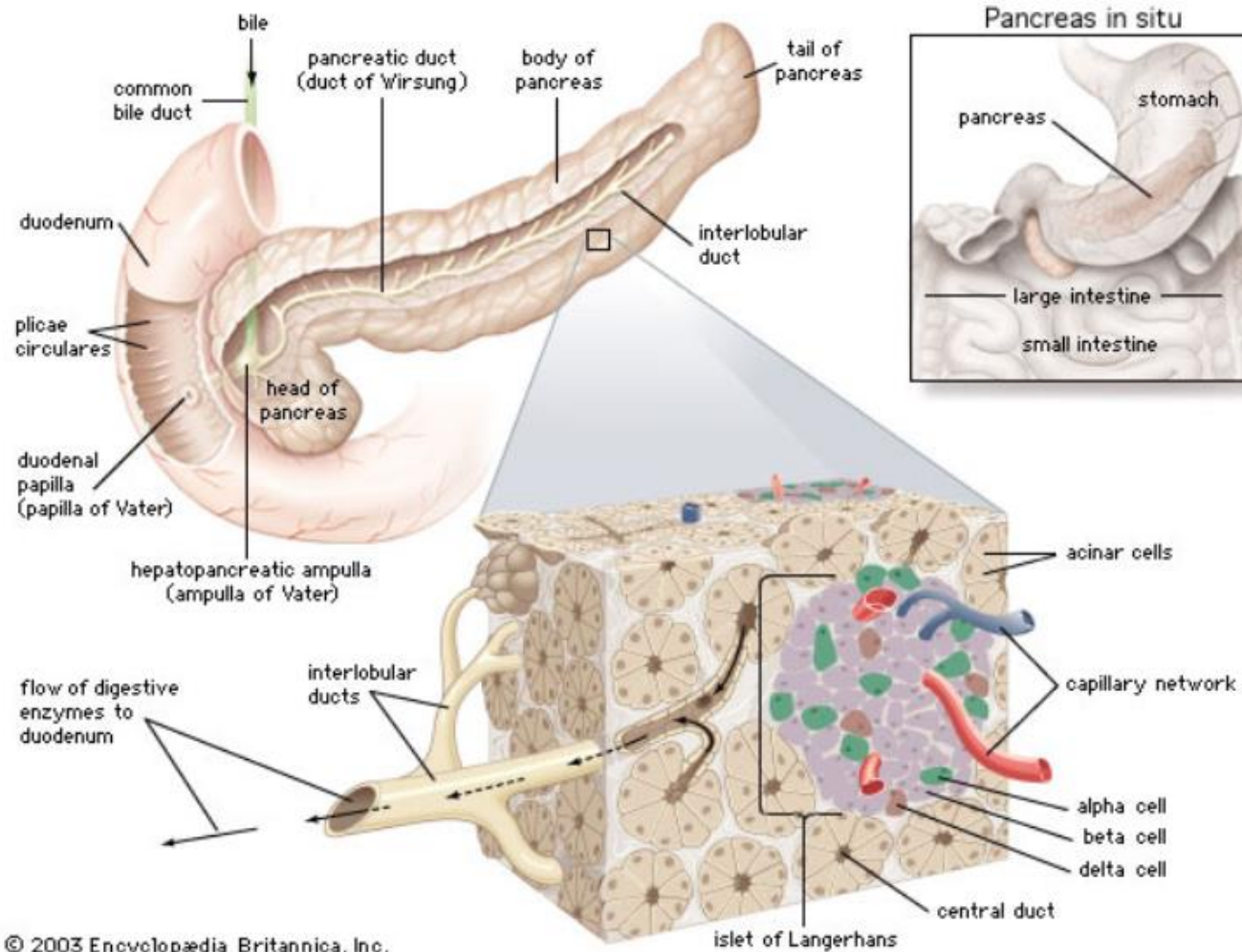
The hormonal response expected as a direct result of a lower-than-normal blood calcium level is

- A. decreased secretion of PTH and increased deposition of calcium in the bones
- B. decreased secretion of calcitonin and increased deposition of calcium in the bones
- ☒ C. increased secretion of PTH and increased release of calcium from the bones
- D. increased secretion of calcitonin and increased release of calcium from the bones

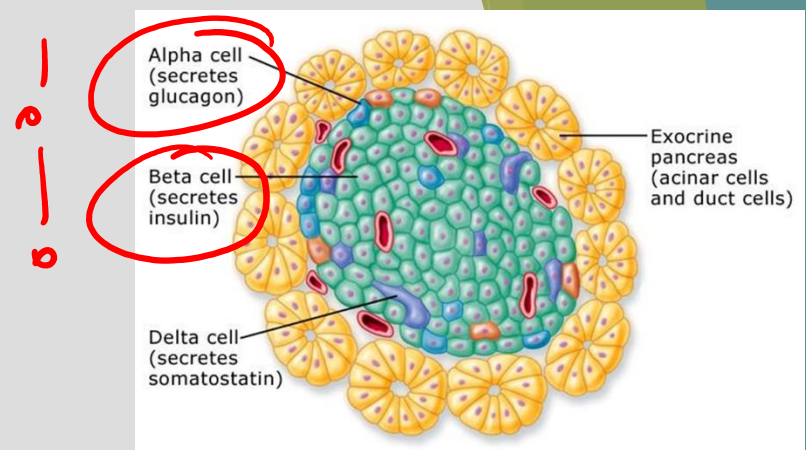
The release of thyroxine from the thyroid gland is directly regulated by

- ☒ A. TSH
- B. PTH
- C. iodine
- D. thyroxine

Pancreas

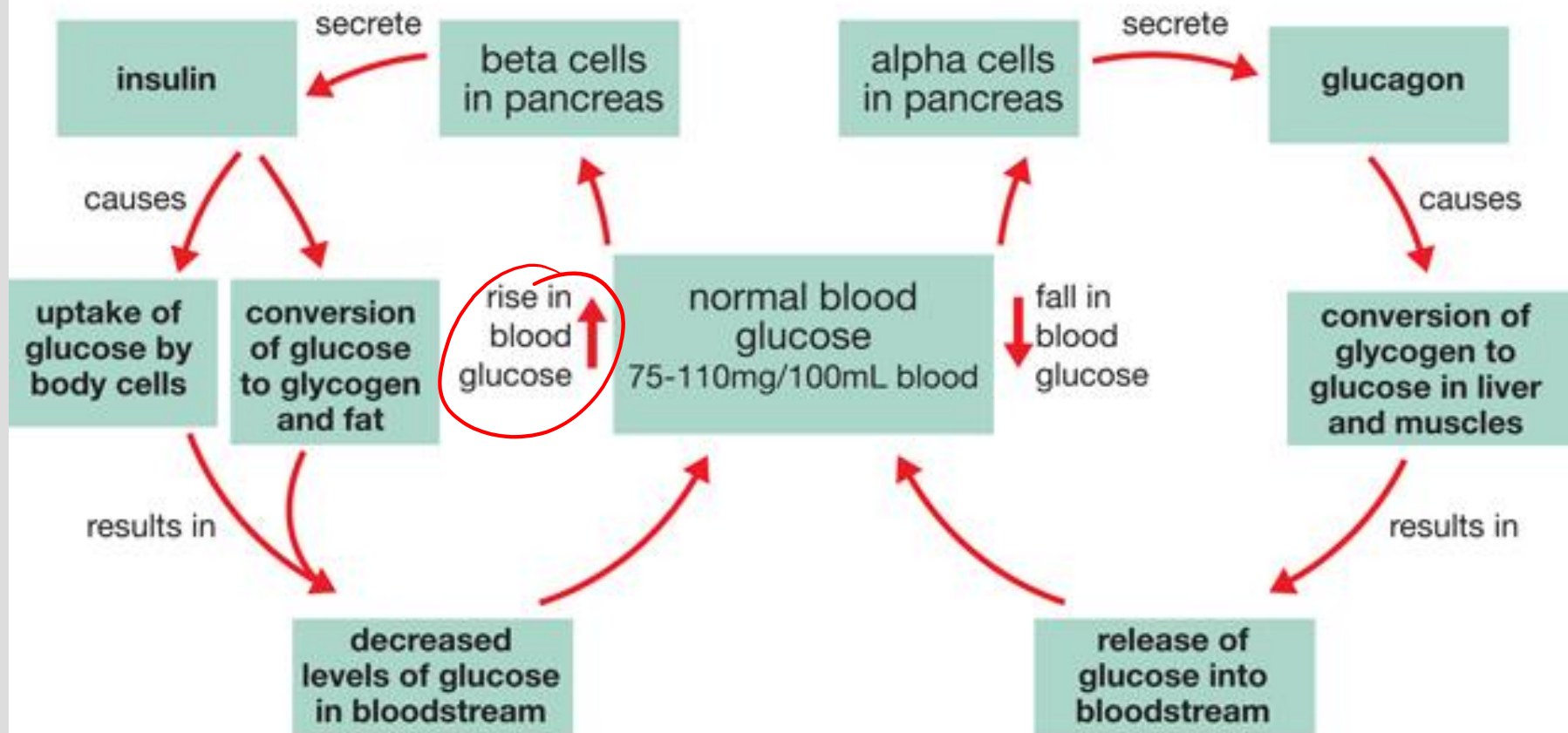


Pancreas



- ▶ hormones of the pancreas act antagonistically to maintain blood glucose:
 - ▶ What is the function of insulin?
 - ▶ Decrease blood glucose levels by:
 - ▶ Increasing permeability of cells to glucose → increased uptake of glucose into cells from blood
 - ▶ What is the function of glucagon?
 - ▶ Increase blood glucose levels by:
 - ▶ Stimulating liver to break down glycogen to release glucose into blood
 - ▶ Activating gluconeogenesis to convert amino acids into glucose
 - ▶ Breaking down fats into fatty acids to be used as fuel

Key to Blood Glucose Regulation Through Negative Feedback



Diabetes Mellitus

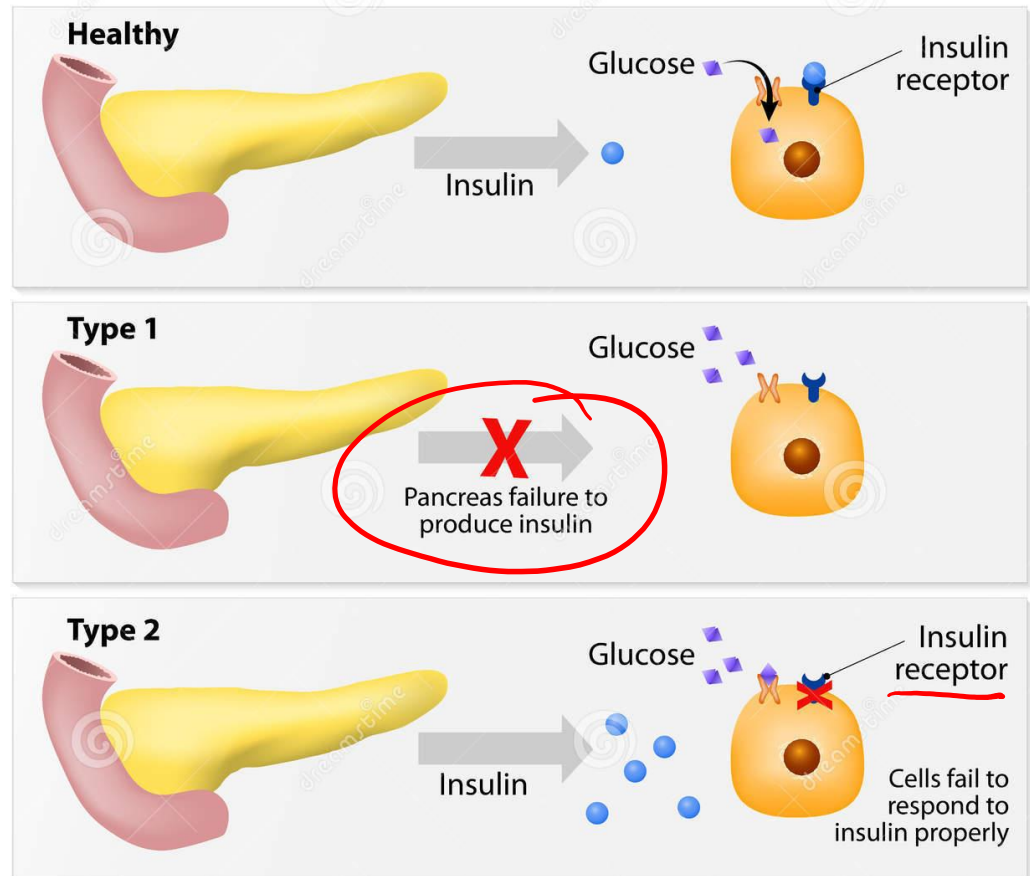
▶ Two types:

- ▶ Type 1 - Juvenile onset
- ▶ Type 2 - Adult onset

▶ Symptoms:

- ▶ Increased thirst, frequent urination, extreme hunger, unexplained weight loss, fatigue.

DIABETES MELLITUS

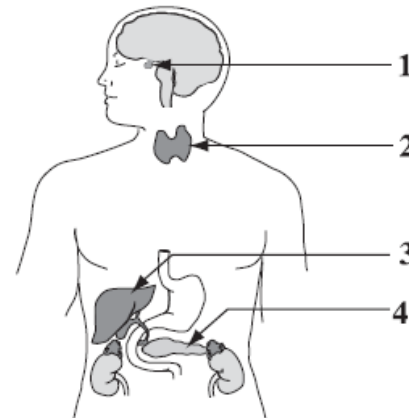


Practice

Use the following information to answer the next two questions.

A drug known as exenatide is used in the treatment of type 2 diabetes. Exenatide stimulates the secretion of insulin and reduces the activity of glucagon.

Some Glands in a Human



8. In the diagram above, the gland that is targeted by exenatide is labelled

- A. 1
- B. 2
- ☒ C. 3
- D. 4

9. Exenatide causes blood glucose levels to

- A. increase by causing an increase in the permeability of cells to glucose
- ☒ B. increase by causing a decrease in the permeability of cells to glucose
- C. decrease by causing an increase in the permeability of cells to glucose
- D. decrease by causing a decrease in the permeability of cells to glucose

Use the following information to answer numerical-response question 6.

Actions of Some Hormones

- 1** Raises the blood glucose level
- 2** Lowers the blood glucose level
- 3** Causes excess glucose to be converted into fat
- 4** Increases permeability of cell membranes to glucose
- 5** Decreases permeability of cell membranes to glucose
- 6** Results in the conversion of glycogen to glucose in the liver
- 7** Results in the conversion of glucose to glycogen in the liver

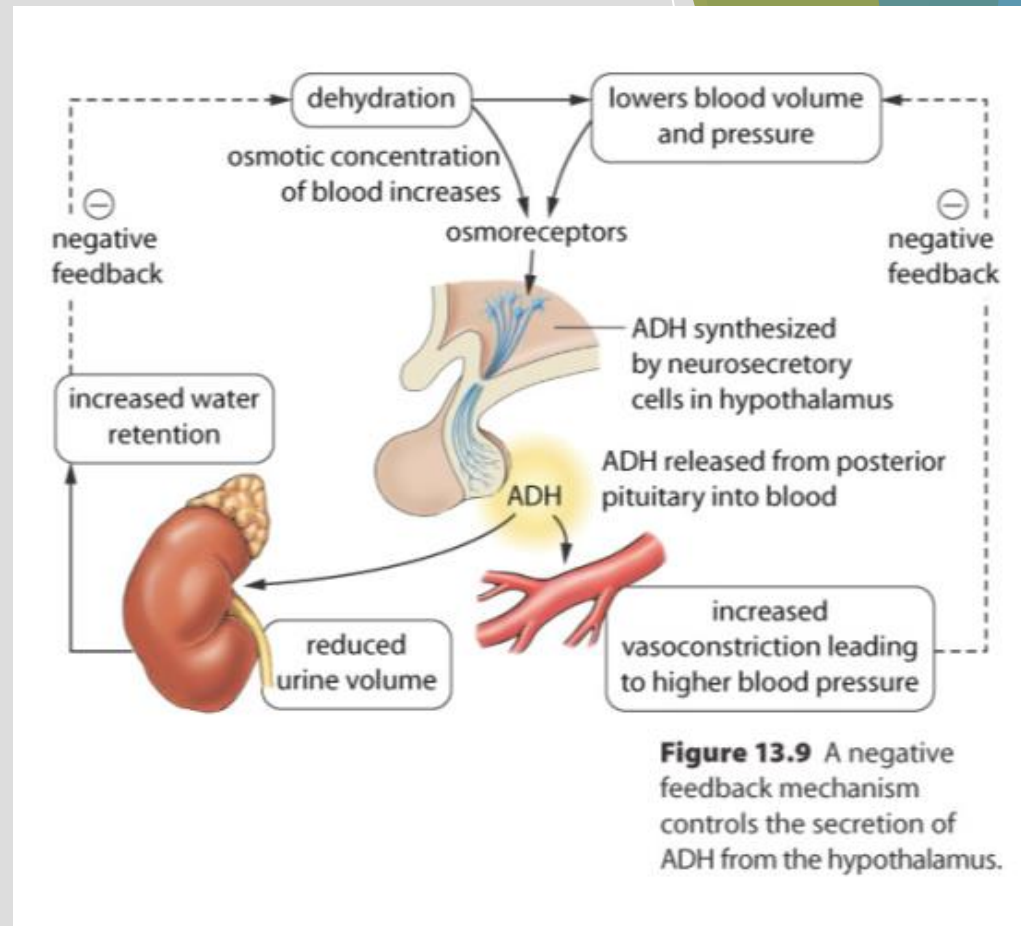
Numerical Response

- 6.** Four actions that describe the functions of insulin are numbered _____, _____, _____, and _____.

(Record all **four digits** of your answer **in any order** in the response area at the bottom of the screen.)

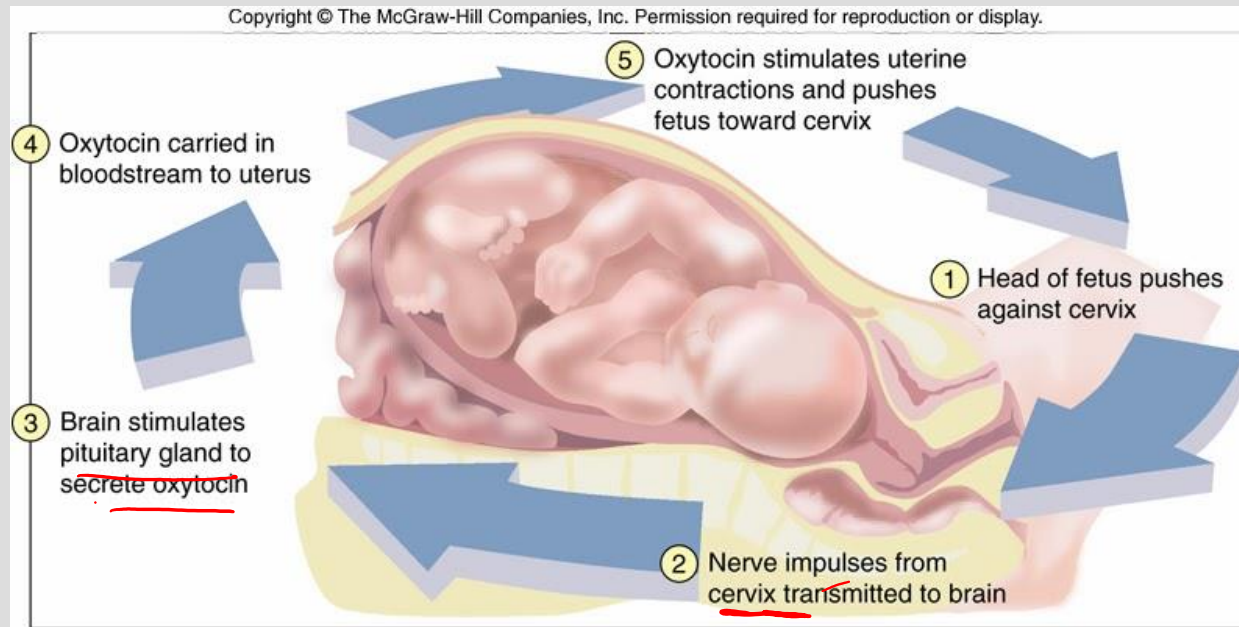
Posterior pituitary gland

- ▶ Antidiuretic hormone (ADH, vasopressin)**
 - ▶ Acts on the collecting duct of the nephron in the kidney, causes you to retain water
 - ▶ Released when dehydrated
 - ▶ Also increases blood pressure (more water in blood)
 - ▶ Lack of ADH causes diabetes insipidus (can't retain water)



Oxytocin → hormone that slows things to be pushed out of the body

- ▶ Stimulates uterine contractions during childbirth (positive feedback)
- ▶ Releases milk from mammary glands
 - ▶ Note: Prolactin allows mammals to produce milk



Practice

Chemicals found in alcohol and tea have a diuretic effect. Diuretics cause the body to produce a greater-than-normal volume of urine.

23. The hormone whose effect is inhibited by diuretics is

- A. TSH
- ☒ B. ADH
- C. insulin
- D. cortisol

Diabetes insipidus is a disorder in which the body fails to produce sufficient levels of ADH.

32. A symptom **most likely** expected in a person who has diabetes insipidus is

- ☒ A. the production of large amounts of dilute urine
- B. a decrease in the glucose concentration in the blood
- C. an increase in the glucose concentration in the urine
- D. the production of small amounts of concentrated urine

Two hormones that work in opposition to each other are

- A. TSH and thyroxine
- ☒ B. insulin and glucagon
- C. ADH and aldosterone
- D. prolactin and oxytocin

Use the following information to answer questions 8 and 9.

Wolfram syndrome is a rare genetic disorder characterized by large amounts of dilute urine, symptoms of type 1 diabetes mellitus, and reduced hearing and vision.

8. Which of the following rows identifies the expected blood-glucose level and the expected blood-insulin level in a person with Wolfram syndrome?

Row	Blood-glucose Level	Blood-insulin Level
☒ A.	Higher than normal	Lower than normal
B.	Higher than normal	Higher than normal
C.	Lower than normal	Lower than normal
D.	Lower than normal	Higher than normal

9. In a person with Wolfram syndrome, the amount and concentration of urine produced could result from
- A. lower-than-normal secretion of ADH from the adrenal gland
 - ☒ B. higher-than-normal secretion of ADH from the adrenal gland
 - C. lower-than-normal secretion of ADH from the pituitary gland
 - D. higher-than-normal secretion of ADH from the pituitary gland